

**SMART LoBo TRAINING KIT: HANDS-ON IOT-BASED LOGIC BOARD
TRAINING KIT FOR ECE STUDENTS AT THE UNIVERSITY OF
RIZAL SYSTEM - MORONG CAMPUS**

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Bachelor of Science in Electronics Engineering

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APPROVAL SHEET

This undergraduate methods of research project entitled SMART LoBo TRAINING KIT: HANDS-ON IOT-BASED LOGIC BOARD TRAINING KIT FOR ECE STUDENTS AT THE UNIVERSITY OF RIZAL SYSTEM - MORONG CAMPUS, prepared and submitted by Ashanty Madeline DC. Cruz, Brian Ezekiel D. Batalon, Mark Jayson F. Villacura, and Tristan Manuel T. Sto. Tomas, in partial fulfillment of the requirements for the degree Bachelor of Science in Electronics Engineering, is hereby recommended for approval.

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-The Researchers

DEDICATION

This study is wholeheartedly dedicated to our beloved parents, who have been our source of inspiration, who believed in us, and gave us strength when we thought of giving up, and who continually provide their moral, spiritual, emotional and financial support. To our siblings, relatives, mentors, and friends who shared their words of advice and who have witnessed all of our hardships and provided encouragement to finish this study. And lastly, we dedicate this piece of hard work to the Almighty God, to whom we have gotten our strength, wisdom, understanding, and inspiration in accomplishing this work.

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ABSTRACT

TITLE	:	SMART LoBo TRAINING KIT: HANDS - ON IoT-BASED LOGIC BOARD TRAINING KIT FOR ECE STUDENTS AT THE UNIVERSITY OF RIZAL SYSTEM - MORONG CAMPUS
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EXECUTIVE SUMMARY:

The general objective of this project is to design and develop a hands-on IoT-based Logic Board Training Kit for Electronics Engineering students at the University of Rizal System-Morong Campus. It focused on creating a user-friendly and accessible tool for students to practice logic circuit problems in two modes: Training Mode and IoT Mode. This innovation supports real-world logic circuit design and facilitates hands-on learning, providing a web-based application for customized problem sets and a training module with discussion and activities. The study was conducted at the University of Rizal System-Morong Campus during the first semester of the 2024-2025 academic year.

The researchers used a developmental approach and conducted a descriptive study to assess the system's acceptability. A questionnaire checklist was created and reviewed before being given to respondents. The respondents of the study were composed of thirteen (13) experts which include ECE graduates and faculty members, and forty-seven (47) third-year ECE students of URS Morong because they are the target end users of the device since they are taking Digital Electronics course. The accuracy of device was determined through comparison of the Logic Simulator Software and using confusion matrix.

The findings are as follows: The Smart LoBo Training Kit: Hands-on IoT-based Logic Board for ECE students at the University of Rizal System-Morong Campus, was successfully designed and developed with the intended purpose of enhancing learning in digital electronics, particularly logic circuits. Moreover, the Smart LoBo Training Kit demonstrated high accuracy in identifying logic outputs, with a 95.68% accuracy rate when compared to the Logisim logic simulator across various difficulty levels (Easy, Average, and Difficult). This was based on the evaluation of nine distinct logic expressions and the use of a confusion matrix, which showed minimal false positives and false negatives. Additionally, a survey conducted with 60 participants, including 13 experts and 47 students, revealed strong approval of the kit's performance. The level of acceptability of the Smart LoBo Training Kit was perceived by the respondents as Very Much Acceptable with an overall average weighted mean of 4.71. Specifically, the level of acceptability of the device in terms of functionality, performance efficiency, usability, reliability, and maintainability received weighted means of 4.78, 4.71,

4.72, 4.62, and 4.73, respectively.

Based on the findings, the researchers concluded that the Smart LoBo Training Kit: Hands-on IoT-based Logic Board for ECE students at the University of Rizal System-Morong Campus is suitable and effective for enhancing the learning experience in digital electronics, specifically logic circuits. The kit performed excellently in its functions and was found to be highly acceptable. It achieved high accuracy rate logic output identification. The training kit received high ratings across key areas, including functional suitability, performance efficiency, usability, reliability, and maintainability, demonstrating its effectiveness as a dependable educational tool.

Based on the findings and conclusions, the following recommendations were drawn up. The researchers suggest using a detachable/rechargeable battery to maximize portability. Additionally, they recommend utilizing a single microcontroller to simplify the circuitry and reduce the system's complexity. Replacing the current LCD with a TFT touch screen display would improve the user interface, eliminating the need for push buttons and overcoming the limitations of the 20x4 LCD in terms of text and display capacity. Finally, the researchers encourage future researchers to continuously improve the device's program to further enhance the efficiency of ECE students' analytical abilities in logic circuit design.

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Chapter 1

BACKGROUND OF PROJECT

This chapter covers the background of the study, objectives of the study, theoretical and conceptual framework, the significance of the study, the scope and limitations of the study, and the definition of terms.

Background of the Study

The risen demand of education in developing countries has seen large number of students in classrooms, lecture halls and laboratories in quest for knowledge and skills according to Mutuku et al. (2020). Hands-on learning is particularly effective for students who struggle with traditional lectures, often referred to as slow learners. Ismael (2023) and Korikana (2020) highlights that these students face challenges in retaining information during passive lectures, which can hinder their academic progress. Providing practical, hands-on experiences helps engage these learners and enhances their understanding.

Moreover, active learning has been shown to significantly improve student performance. In a study conducted by Freeman et al. (2014), they found out that active learning reduces failure rates by 55 percent and improves grades by half a letter compared to traditional lectures. The President's Council of Advisors on Science and Technology has recommended a 33 percent increase in STEM degree completions, advocating for active learning as a key strategy. Additionally, Abante et al. (2014) emphasized that kinesthetic or physical learners benefit greatly from hands-on learning, further supporting its effectiveness in improving academic outcomes.

The lack of adequate laboratory equipment is also a problem in education system. In the context of the Philippines, the lack of laboratory equipment educational institution has been a significant challenge. According to Abas and Marasigan (2020), science laboratories are essential for inquiry-based learning, but their study in public junior high schools in Lanao del Sur reveals that inadequate facilities, defective tools, and missing resources like electricity and water hinder effective teaching. In addition, Jaunico et al. (2021) highlight the struggles of science teachers, who experience emotional stress, including guilt and pressure, due to insufficient equipment. As a result, teachers often resort to improvisation, using localized materials, or integrating technology to overcome these limitations. Despite their resourcefulness, the ongoing shortage of laboratory resources remains a barrier to quality science education in the country, emphasizing the need for improvements in laboratory provision to support effective teaching and learning. Bocheng et al. (2014) emphasized integrating practical applications into digital logic education through a network teaching platform, which significantly enhances teaching and learning experiences

Furthermore, access to modern laboratory equipment in Philippine schools is limited, with only 15 percent of schools having digital devices, as reported by Pingol et al. (2020). Their Versatile Instrumentation System for Science Education and Research (VISSER) aims to provide affordable, versatile lab tools for various sciences. Timbasal-Nuevo (2024) found that inadequate equipment affects the performance of junior high school science teachers in Malapatan Municipality, revealing a need for improved teacher training to better utilize available resources

and enhance teaching effectiveness. These findings further emphasize the importance of improving access to modern and adequate laboratory equipment, alongside investing in teacher development, to ensure effective science education in the country.

Moreover, Tomaro (2018) discussed the integration of Information and Communication Technology (ICT) in the Philippine educational system, revealing both progress and persistent challenges. The study emphasized the need for strategic adjustments in the curriculum, enhanced teacher training, and improved infrastructure. Despite governmental efforts to prioritize ICT, inadequacies in school facilities continue to hinder effective implementation, affecting the country's technological advancement and global competitiveness.

To supplement conventional teaching methods, use of new technology have recently been adapted in the current education system. Such examples are implementing simulators and quiz kits in a classroom session to assist teachers in providing standardized evaluation or practical and more direct examples of concepts that are being taught, based on the study of Mohammad et. al (2019). Hence, the researchers came up with an idea to provide enhancement and modification in learning management in specific subject which is digital logic circuit.

Digital electronics plays a crucial role in today's era as technology widely used digital electronic system. Timbol (2019) defines logic circuits as the fundamental building blocks of digital systems. These circuits process binary data, which is represented by two states: on (1) and off (0). Logic gates, which are part of these circuits, perform basic logical operations on binary inputs. Common types

of logic gates include AND, OR, and NOT, each of which produces a specific output based on the combination of its inputs. Logic circuits are used in various devices, from simple systems with a few gates to complex systems like microprocessors that may contain millions of gates. These circuits are essential for the functioning of digital devices, enabling them to process and store binary information.

In this context, Boolean expressions are integral to the design and understanding of logic circuits, as they simplify complex digital systems into more manageable forms. Hamed (2019) explains that Boolean algebra plays a crucial role in defining and optimizing logic circuits, particularly in control and protection circuits. By using Boolean equations, which are derived from binary values (1 and 0), digital logic gates such as AND, OR, and NOT can be represented and analyzed. The process begins with the use of truth tables, which are systematically derived from Boolean expressions to show the relationship between inputs and outputs.

Similarly, logic gates are the most important components in utilizing logic circuits and Boolean expressions. Garg (2019) elaborates the importance of logic gates, discussing how different types such as AND, NOT, OR, NAND, NOR, XOR, and XNOR perform specific logical operations. Garg also highlights the application of these gates in real-life systems, such as burglar alarms and security systems, and notes how modeling physical systems (e.g., mechanical, optical, and electrical) can be accomplished using Boolean expressions, which are integral to logic gate operations. Together, these studies emphasize the role of logic circuits in digital systems, making them crucial for decision-making and information

processing in modern technology.

Moreover, functions of logic gates are embedded in electronic circuits, specifically in Integrated Circuits (IC). According to Jin (2023), the history of ICs traces significant advancements, starting from the discovery of semiconductor properties to the invention of the first transistor, microprocessor, and the development of integrated circuit technologies. Hào (2024) further emphasizes that digital ICs utilize these logic gates to manage binary signals, allowing essential functions such as data storage, instruction processing, and communication between devices. Thus, the integration of logic gates within ICs is central to the functionality and advancement of electronic systems. Also, he discussed that logic gates, such as AND, OR, NOT, and XOR, are fundamental to the operation of digital ICs, enabling binary computations crucial for processing information. Farhan et al. (2023) created a training board that effectively links theoretical concepts with practical applications, employing switches, diodes, transistors, and integrated circuits to provide a comprehensive educational tool.

In the context of education, these logic circuits are being discussed not only for their applications but also as key elements in modern teaching. According to Bakeyeva et al. (2020), the digitalization of education has shifted the focus toward fostering independent, creative, and responsible learners. Structural and logical schemes, such as logic circuits, have become key components in this digital transformation, as they help students understand complex concepts by organizing information into clear, logical structures. This approach not only enhances the students' cognitive abilities but also promotes engagement through modern

educational technologies.

Building on this, the use of technology-based tools for teaching logic circuits has proven to be effective in the Philippines. A study by Arante (2018) at Caraga State University demonstrated the effectiveness of an improvised Logic Gates Simulator as a teaching aid in basic digital electronics. Kaur et al. (2018) reviewed dynamic CMOS logic circuits, highlighting their use in high-speed digital processors. The study found that students who learned using the simulator performed significantly better than those taught through traditional methods. This indicates that hands-on learning with simulated logic operations can improve student performance, making it a valuable tool in electronics technology courses. The study highlights the importance of integrating practical simulations to reinforce theoretical knowledge and enhance the learning experience for students in the Philippines.

As a result, the use of training kits in education has become increasingly valuable for enhancing learning management. According to Serafin (2018), electro-technical assembly kits serve as important tools in technical education by reinforcing both practical and theoretical understanding among students. These kits play a dual role in education, where they not only aid instruction but also become a key part of the learning curriculum. Furthermore, Serafin notes the increasing relevance of connecting these kits with computers to meet evolving educational standards, thus enhancing their value in technical training. Likewise, Jambari et al. (2018) observe that Educational Trainer Kits (ETK) significantly improve teamwork and problem-solving abilities among engineering students, with

over 95.5 percent of participants affirming the effectiveness of ETKs. Their research highlights the importance of incorporating modern technology in training kits to strengthen technical knowledge and skills, making ETKs essential for effective technical education. These studies show how helpful educational training kits are. Therefore, conducting a study such as logic board training kit will be helpful to ECE students.

In addition to prior studies, the existence of logic training kits makes technical education more accessible and engaging for students. Mohammad et al. (2019) developed the e-Logic Trainer Kit (e-LTK), an interactive educational tool that supports traditional teaching methods with a simulator and quiz module focused on logic gate circuits. Using an Arduino-based interface with a keypad and display, the e-LTK allows students to test different logic gate combinations and view timing diagrams, which strengthens their understanding of circuit functions. Positive feedback from students and instructors at Universiti Teknikal Malaysia Melaka, with 60 percent indicating favorable results, underscores the effectiveness of such training kits in enhancing technical learning experiences. Universiti Teknikal Malaysia Melaka's students and instructors demonstrated its effectiveness, with 60 percent reporting favorable responses, thereby underscoring the practical value of such training kits in technical education. Similarly, Khaing et al. (2018) developed a compact and portable Arduino-based logic gate training kit for university engineering students. This kit includes input selectors, LED indicators, and battery power, allowing users to explore six fundamental logic gates (AND, NAND, OR, NOR, XOR, XNOR) and immediately

observe outputs. These tools encourage active learning and reinforce key concepts in technical education, with both studies demonstrating positive responses from students, underscoring the effectiveness of logic training kits in the classroom.

The researchers developed a study designed to enhance students' hands-on skills in digital electronics, specifically focusing on real-life simulations of logic circuits with basic Internet of Things (IoT) applications. The purpose of this study aligns with the objectives set forth in Republic Act No. 10844, also known as the Department of Information and Communications Technology (DICT) Act of 2015, which aims to establish the Department of Information and Communications Technology, define its powers and functions, and appropriate necessary funds. The law seeks to foster technological advancements, including in education, to enhance students' knowledge and skills.

As stated in Section 2, Declaration of Policies, paragraph (h) of Republic Act No. 10844, the State aims:

“To promote the use of ICT for the enhancement of key public services, such as education, public health and safety, revenue generation, and socio-civic purposes”

This section underscores the importance of utilizing ICT not only to modernize public services but also to advance educational systems. By incorporating advanced technologies like IoT and digital electronics into educational training, this study seeks to equip students with practical, real-world skills that are vital in today's technology-driven environment.

The researchers also align their study with the policy objectives of Republic

Act No. 11293, also known as the Philippine Innovation Act, which recognizes innovation as a critical component of the country's development strategies. The act also allocates funding to foster innovation, science, and technology, and aims to enhance the country's overall innovation ecosystem.

As specified in Section 2, Declaration of Policies, letter (c) of Republic Act No. 11293:

“Investments in education, science, technology and innovation shall be guided by a strategic direction towards strengthening the country's knowledge-based economic development that benefits all. Innovation starts with a robust pool of skilled, talented and creative people. Educational institutions, private organizations, government agencies and local government units (LGUs) are key drivers of programs that stimulate innovation literacy and skills development for the Filipino workforce and entrepreneurs, including women and the youth”

This section emphasizes that the development of a skilled workforce, supported by investments in education and innovation, is essential for fostering a knowledge-based economy that benefits all sectors of society. By aligning with this policy, the researchers ensure that the Smart LoBo Training Kit is an investment in education that directly supports the development of students' practical skills in the areas of digital electronics and IoT. This device contributes to the laboratory activities within the university, equipping students with the necessary skills and knowledge to drive innovation in their future careers.

In summary, this study highlights the importance of hands-on, active learning in supporting student success, especially in developing countries like the Philippines, where limited resources often impede educational outcomes. In the context of the Philippines, the scarcity of laboratory equipment forces teachers to improvise, affecting education quality. Technological tools like simulators and

training kits have shown effectiveness in building practical skills, particularly in technical areas like digital electronics. To address this gap, the researchers propose the Smart LoBo Training Kit: An IoT-Based Logic Board for ECE students at the University of Rizal System–Morong Campus. This project seeks to provide a practical, cost-effective solution to improve hands-on training in digital electronics. Moreover, the development of this training kit aligns with key policies set by the Philippine government, particularly those related to Information and Communications Technology (ICT) and educational innovation. By integrating advanced technologies into the curriculum, the project supports the growth of students' skills in line with national goals for technological advancement and career preparedness.

Objectives of the Study

Generally, this project aims to develop an interactive educational tool, a Hands-on IoT-Based Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus to enhance their practical skills and knowledge in digital electronics.

Specifically, this project sought to:

1. Design an IoT-Based Logic Board Training Kit with the following features:
 - 1.1 Web-based Application;
 - 1.2 User Manual;
 - 1.3 Portability;
 - 1.4 User-friendly Accessible Design; and
 - 1.5 Logic Gates and Boolean Algebra Training Module which consists of:

1.5.1 Discussion Content; and

1.5.2 Activities.

2. Evaluate the system performance of the device in identifying the logic output in terms of accuracy.
3. Determine the level of acceptability of the design project as a reference to ISO/IEC 25010:
 - 3.1 Functional Suitability;
 - 3.2 Performance Efficiency;
 - 3.3 Usability;
 - 3.4 Reliability; and
 - 3.5 Maintainability.

Theoretical Framework

The theories and principles of this study are anchored on the Laws of Boolean Algebra and Switching Theory as cited by Storr (2021). These are essential for understanding the operation of logic circuits in digital electronics.

Boolean Algebra is one of the mathematical systems that uses only two variables whose values are true and false or 1 and 0 respectively. This concept points out to the Laws of Boolean Algebra which include a list of propositions that help reduce the number of logic gates needed to perform a given logical operation. There are different ways to perform logic operations, these include the Identity Law, Idempotent Law, Complement Law, and De Morgan's Theorems which are important in simplifying logic expressions.

Upon developing the Logic Training Kit, it is essential to apply the Laws of

Boolean Algebra as it helps student improve their understanding of the concepts of logical analysis. It will enable students to utilise different Boolean theorems or techniques to create and solve complex logic circuits. Such simplification leads to better knowledge and practical skills in the field of electronics.

Switching Theory is related to the concept of Boolean Algebra. The logic circuit consists of a switch, typically a transistor that can change voltage to high and low or 1 and 0 respectively. The idea of Switching Theory is that it can be used to implement Boolean expressions and digital logic gates that perform various operations based on their switch configurations, including NOT, AND, OR, NOR, and NAND gates.

This Switching theory enables students to understand how logic gates work. It also provides an idea for creating different switch algorithms for logic circuits, including those used in Programmable Logic Controllers (PLCs). This foundational knowledge is crucial for developing practical skills in electronics.

Conceptual Framework

The conceptual framework of this study, as shown in Figure 1, is based on System Development Life Cycle (SDLC) Model a systematic approach to instructional design. This model ensures effective development and evaluation of the training kit for Electronics Engineering students at the University of Rizal System-Morong Campus.

Firstly, the analysis is conducted, to develop Smart LoBo Training Kit for target audience. Information is gathered from various sources, including related studies on logic circuit design, articles, and online resources.

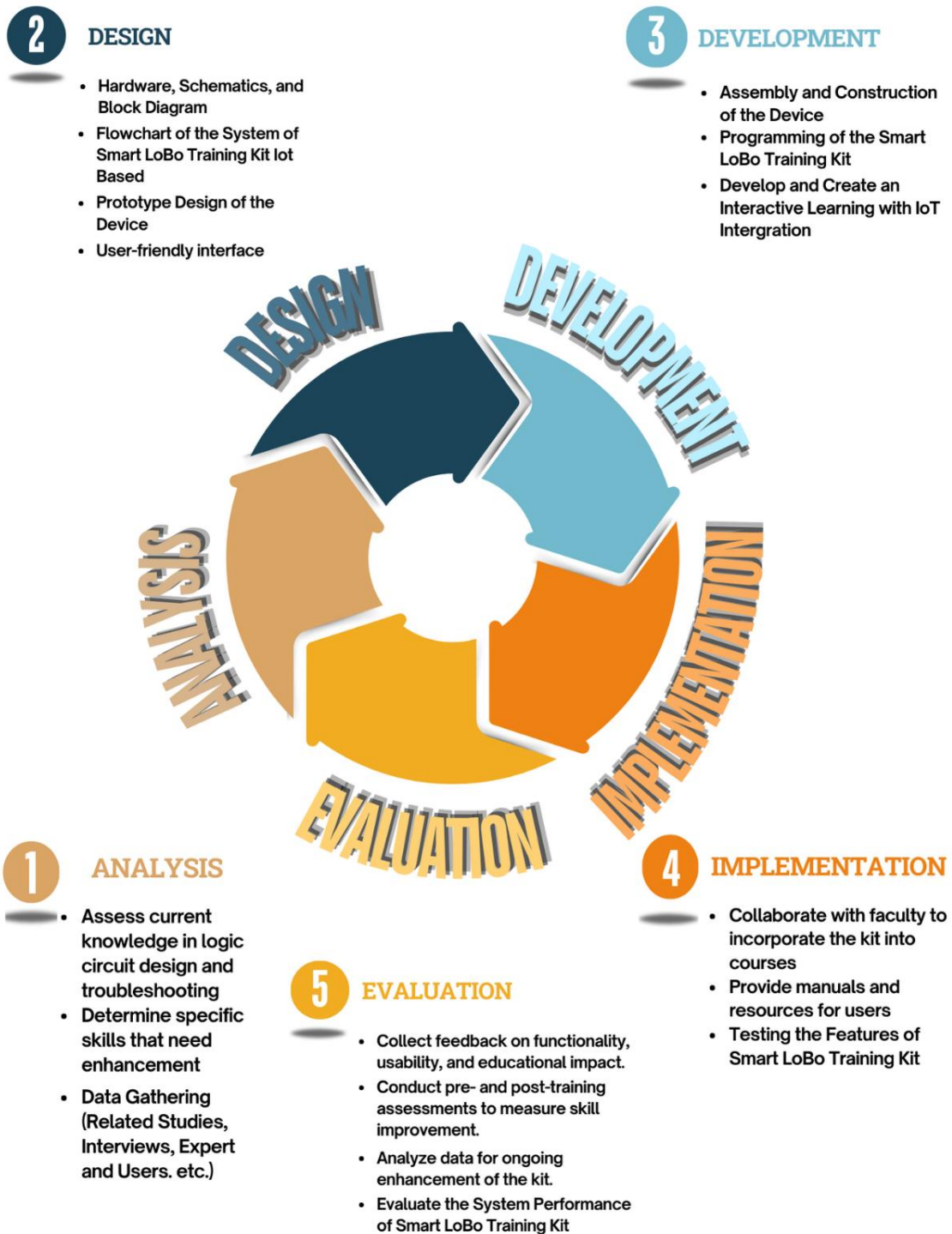


Figure 1

Modified Conceptual Framework of Smart LoBo Training Kit:
Hands-on Logic Board Training Kit for ECE Students at
the University of Rizal System-Morong Campus

These sources provide insights into the current skill levels and areas for improvement of students, engaging with instructor and potential users through meeting that plays a significant role in understanding practical applications and developing the training kit to ensure that project met the study's objectives.

During the analysis stage, clear objectives are followed, focusing on improving practical skills in logic circuit design involving the existing curriculum that includes detailed discussions on logic circuit and a modular design for easy assembly. Designing stage, his phase which a prototype design of the device, flowchart of the system of Smart LoBo Training Kit and user-friendly interface. Next is the development stage, this includes the assembly and construction of the device, programming of software of web-based application. While the implementation stage provides instructional materials, such as user manual and user guides. Initial testing of the features and performance to gather user feedback, allowing for adjustments to enhance usability.

Finally, in evaluation stage through collaboration with faculty to introduce the training kit to ECE students, ensuring effective integration into existing coursework. This stage is conducted through various assessments, including accuracy testing of logic output identification and user satisfaction surveys. The feedback collected during this process informs improvements, ensuring that the Smart LoBo Training Kit provides a meaningful learning experience for students. Ultimately, the goal of this project is to equip students with essential hands-on skills in logic circuit design, fostering their educational development and readiness for the field.

Significance of the Study

This study aims to develop Smart LoBo Kit: Hands-on IoT-based Logic Board Training Kit for ECE Students at University of Rizal System-Morong Campus,

The researcher's study offers significant benefits to the following:

ECE Faculty of the University of Rizal System-Morong Campus. This study would help the ECE faculty at the University of Rizal System-Morong Campus to enhance learning management for the ECE students.

ECE Students of the University of Rizal System-Morong Campus. This study will help them to improve and contributes academic exercises on enhancing their learning ability.

Future Researchers. This study helps them to innovate the Smart LoBo Kit: Hands-on IoT-based Logic Board Training Kit for ECE Students at University of Rizal System-Morong Campus. Also, this study will serve as technological advancement for student learning management.

Scope and Limitations

This study, entitled Smart LoBo Kit: Hands-on IoT-based Logic Board Training Kit for ECE Students at University of Rizal System-Morong Campus, was conducted to design and develop an IoT-integrated logic board training kit to improve the practical learning experience among the Electronics and Communication Engineering (ECE) students at the University of Rizal System, located on Sumulong Street, Morong, Rizal. The research was conducted during

the first semester of the academic year 2024-2025.

The study utilized a developmental research approach, focused on the creation an IoT-enabled logic training kit specifically for digital logic courses that executes the real-word logic circuit design. Additionally, a descriptive research method is employed to evaluate the training kit's acceptability based on ISO/IEC 25010 quality standards, using a Likert scale questionnaire administered to 47 ECE students and 13 ECE graduates and professors. The responses from this questionnaire provided insights into the training kit's functional suitability, performance efficiency, usability, reliability, and maintainability.

The Smart LoBo Training Kit training kit aims to enhance students' hands-on learning in logic circuit design through features such as portability, a user-friendly design, and real-time implementation of logic circuits. The training kit also includes a web-based application for the customize problem sets in IoT mode where professors can upload problem sets to support students' learning, a user manual, and a logic gates and Boolean algebra training module that provides both instructional content and activities. Although the kit includes interactive features and allows instructors to set customized problem sets, real-time monitoring of students' actions is not available.

Furthermore, the number of available logic blocks is limited to a total of thirty-two (32) blocks. Specifically, the kit provides five logic blocks each for OR, AND, NOT, NOR, NAND, and XOR gates, and only two XNOR gates. This limitation may restrict the range of logic expressions that users can implement on the logic board trainer.

This study focused on evaluating the accuracy and acceptability of the training kit within the curriculum of digital logic, excluding its interoperability with external systems and applicability to other courses. The findings are specific to the University of Rizal System-Morong Campus and are not intended to be generalized to other educational institutions.

Definition of Terms

The important terms used in this study have been defined operationally for clarification.

Boolean Algebra. It is a mathematical system that operates with binary variables (true and false or 1 and 0), providing a framework for expressing and simplifying logical operations through laws and theorems.

Boolean Theorems. These are Principles in Boolean algebra that is used to simplify and manipulate Boolean expressions. Some of the theorems are the Identity Law, Idempotent Law the Complement Law and De Morgan's Theorems that learners can apply when solving complicated logic issues.

Circuit Simulation. It is a method in which digital computer programs simulate the behavior of the circuits on a piece of paper without constructing it physically. It makes it possible for an individual to assess output of the circuit to reduce unnecessary complexities.

Functional Suitability. This refers to the capability of the SMART LoBo Training Kit to provide functions that fulfill the educational and practical requirements of ECE students. This encompasses the accuracy and completeness of the kit's logic circuit simulations and training exercises, ensuring alignment with the academic needs of the University of Rizal System - Morong Campus.

Internet of Things (IoT). It is a network of interconnected devices that communicate and exchange data over the internet. In this study, IoT technology is integrated into the logic training Kit to enhance learning process and provide versatile functionalities.

Integrated Circuit (IC) Board. It is a semiconductor device that integrates multiple electronic components, such as transistors and resistors, into a single chip. This makes it possible to provide complex functionality in logic circuits while minimizing space and power consumption.

Logic Gates. These are basic components of digital circuits that perform logical operations on binary inputs to produce a single binary output. Common types include NOT, AND, OR, NOR, and NAND gates.

Logic Circuit Analysis. It is the process of testing the circuitry of a logic system to identify its response and performance given specific inputs and expected output.

Maintainability. It refers to the ease with which the SMART LoBo Training Kit can be maintained and serviced. This includes the ability to update software, repair or replace hardware components, and address any issues efficiently, ensuring that the kit remains functional and up-to-date for continuous use in training and academic applications.

Performance Efficiency. It refers to the ability of the SMART LoBo Training Kit to deliver fast and optimized performance. This includes efficient use of hardware and software resources, minimizing delays in processing logic operations, and ensuring seamless IoT connectivity for a smooth user experience.

Reliability. It refers to the capability of the SMART LoBo Training Kit to perform consistently and dependably over time. It ensures stable operation throughout training sessions, with mechanisms in place to handle errors and prevent system failures, thereby supporting uninterrupted educational activities.

Truth Table. This table is used to represent all possible input combinations and

corresponding output values for a logical function, essential for logic circuit analysis and design.

Usability. It refers to the degree to which the SMART LoBo Training Kit is easy for ECE students to operate and understand. This involves the clarity of instructions, user-friendly interface, intuitive navigation of modes, and overall design that supports effective hands-on learning and engagement.

Chapter 2

DESIGN METHODOLOGY

This chapter discusses the research technical design, the research instruments, the procedures and methods of the study, the project design models, and the statistical treatment that is used for this study.

Research Technical Design

For the study entitled Smart Lobo Training Kit: Hands-On IoT-Based Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus, the descriptive and developmental methods of research were utilized to achieve the objectives of the study.

According to Pimentel (2018), descriptive research is defined as a research method used to describe the existing phenomena as accurately as possible.

Ibrahim (2016) defined the developmental research as an interactive, cyclic process of development and research in which theoretical ideas of the designer feed the development of products that are tested in classroom settings. This process eventually leads to products that are both theoretically and empirically grounded, enhancing the developer's learning and contributing to local instructional theory.

The technical design of the project was illustrated through several key figures, including the block diagram of the device and the flow charts for both the training mode and IoT mode of the device.

Figure 2 illustrates the system's block diagram, which includes several components: The ESP8266, serving as the communication module for the IoT

mode, and the ESP32, which served as the main board for preset problems in the training mode of the device. A timer module, built into or programmed into the main board, acted as a measuring tool for the students' time spent practicing logic problems. When the timer stopped, a buzzer sounded to indicate that the time was finished. The LCD displayed the set of problems, including the countdown of the timer.

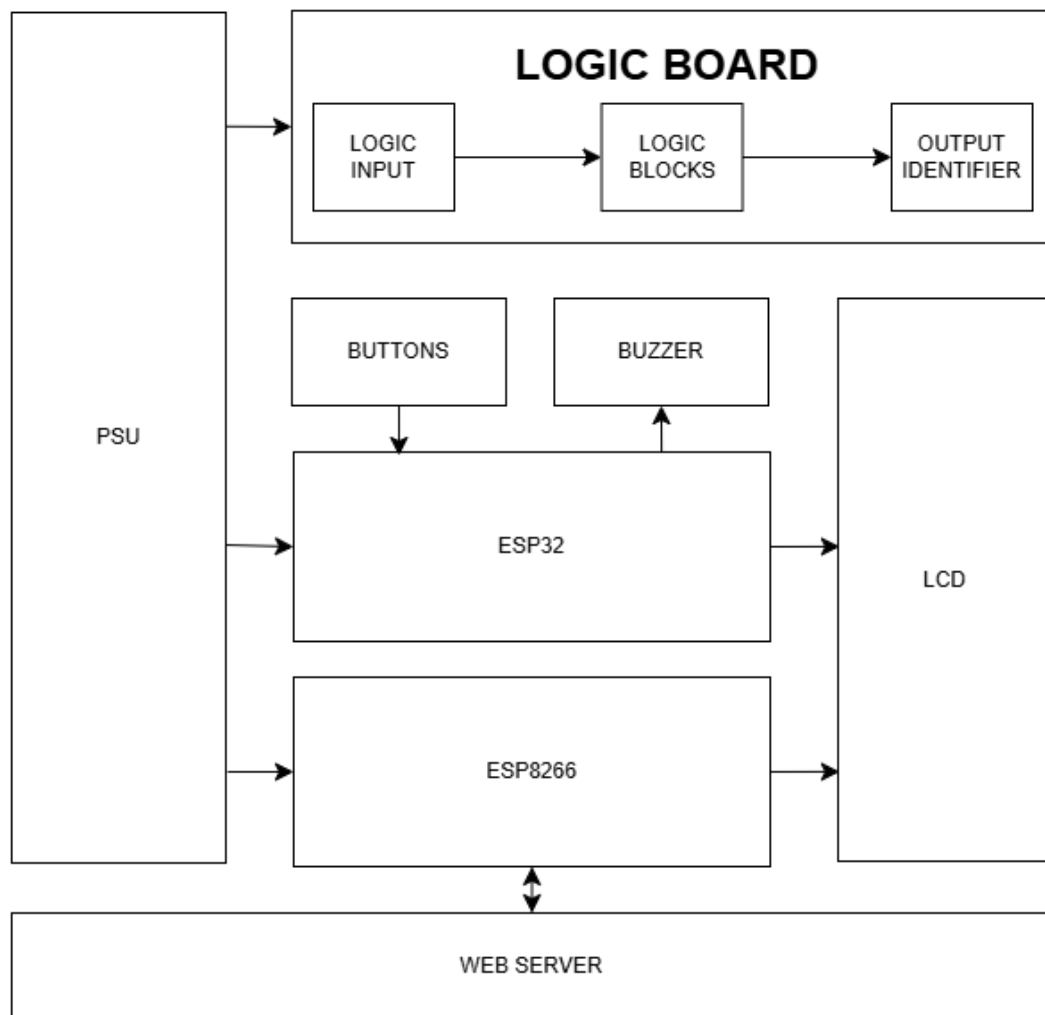


Figure 2

Block Diagram of Smart LoBo Training Kit: Hands-on IoT-Based Logic Board Training Kit for ECE Students at the University Rizal System-Morong Campus

The mode selection is displayed on the LCD, controlled through buttons. Three buttons act as the interface control for mode selection. Both the microcontrollers and logic board were powered, with the logic input consisting of three pins per input: VCC, 0/1 Input, and GND, which pass through the logic gate ICs. VCC and GND also power all ICs placed in the logic board. Finally, the Logic Output Identifier consists of two components: Binary-Coded Decimal (BCD), which determines whether the output is High (1) or Low (0), and this output is displayed on the 7-segment display.

Figure 3 shows the flowchart for the training mode of the device. After initialization, the user can select a mode either Training Mode or IoT Mode. Training Mode includes Practice Mode, where the user could choose the level of difficulty: Easy, Average, or Difficult. Easy involves simpler problems with a 3-minute timer, Average includes moderately difficult problems with a 5-minute timer, and Difficult presents more challenging problems with a 7-minute timer. When the timer stops, the buzzer sounds to notify the user that time is up, and the correct answer is displayed on the LCD. The device then prompts the user to decide whether to try again. If the user selects Yes, the program returns to the selection of difficulty level. If the user selects No, the device will end and resets its function.

Each difficulty level contains five (5) sets of randomized questions, meaning that if the user selects the same difficulty level again, the same set of questions may or may not appear again due to its unique program of being randomized questions. This feature adds variety and helps reinforce the user's learning by exposing them to different logic problems in each session.

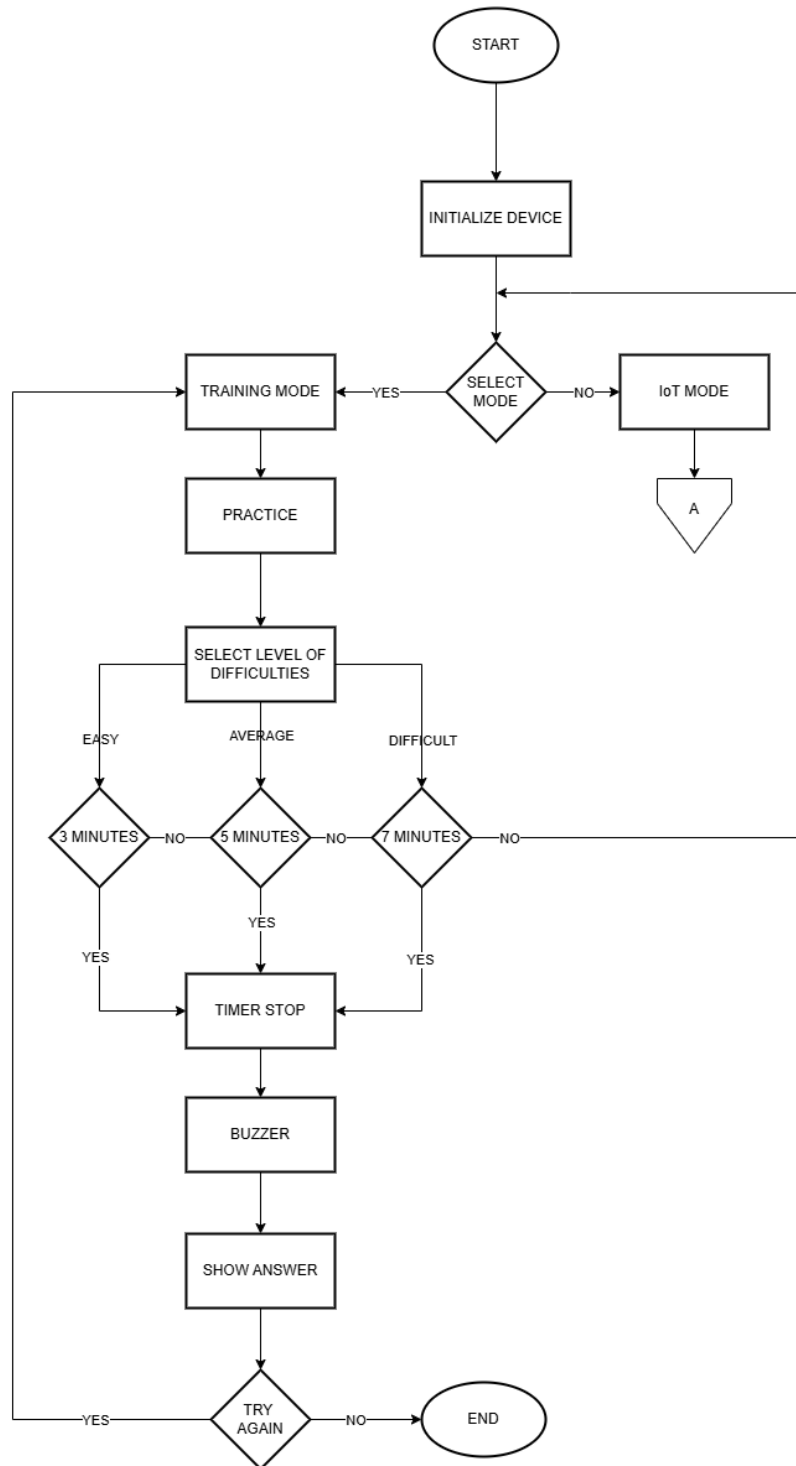


Figure 3

Flowchart for Training Mode of Smart LoBo Training Kit: Hands-On IoT-Based Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus

Figure 4 shows the flowchart for the IoT Mode of the device. The device pairs with the user's Wi-Fi via the ESP8266 module by connecting to the microcontroller's Service Set Identifier (SSID) named Smart Logic Board. The user can then search for the specific Internet Protocol (IP) address 192.168.4.1 in a browser tab. Afterward, the user is directed to a web-application where they can set the number of questions. Once the number of questions was set, the interface displays textboxes that allow the user to input problems, set the timer, and input the answers. The problems are displayed on the LCD, and the timer begins. When the timer stops, the answer is shown. Upon completing the function, the web server displays a "Want to try again?" button, which, when clicked, returns the user to setting the number of questions.

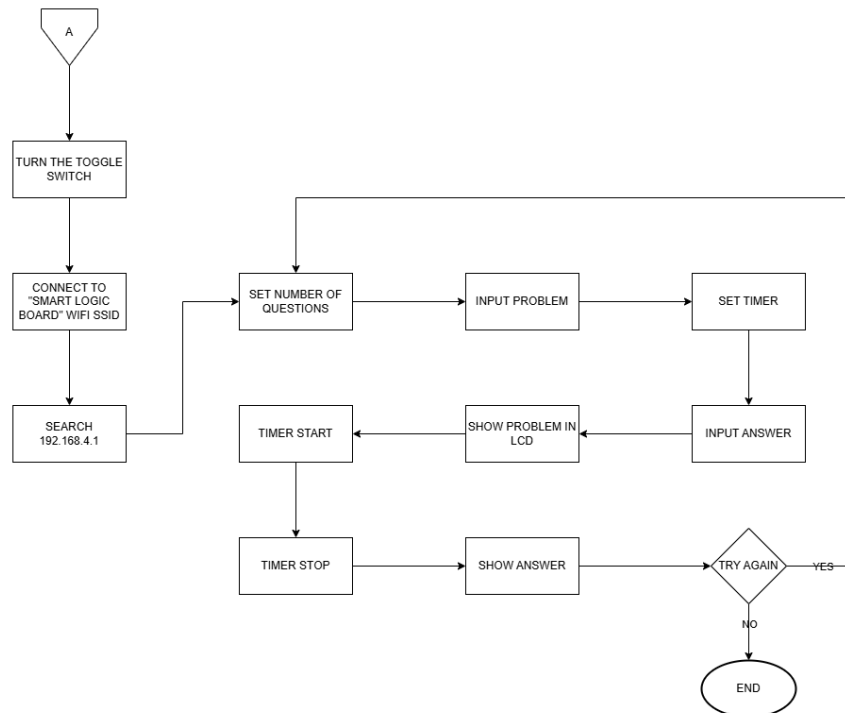


Figure 4

Flowchart for IoT Mode of Smart LoBo Training Kit: Hands-On IoT-Based Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus

Research Instrument

Sugiyono (2018) define a research instrument as an assessment instrument which is useful in measuring observable natural and social phenomena.

For the purpose of developing the logic board training kit, this study used Microsoft Visual Studio Code as the Integrated Development Environment (IDE) for writing and debugging the code. The microcontroller's programming language used was C++ since it is compatible with most of the IoT libraries. Other parts such as the LCD display, timer and buzzer were added to offer further functionality in the project.

A survey questionnaire was developed to capture the user's perception on the design features of the device that are aligned with the objectives of the project. The survey questionnaire results were employed in determining the level of acceptability of the device in terms of functional suitability, performance efficiency, usability, reliability, and maintainability.

The responses were assessed using a Likert scale to gauge the level of acceptability of the device. The Likert scale was structured as follows:

Scale	Range	Verbal Interpretation
5	4.20-5.00	Very Much Acceptable
4	3.40-4.19	Very Acceptable
3	2.60-3.39	Acceptable
2	1.80-2.59	Slightly Acceptable
1	1.00-1.79	Not Acceptable

This scale measures the extent to which the system addresses the needs

of the target groups.

Procedures and Methods of the Study

This section describes the activities carried out in order to accomplish the research. It consists of two parts: Conduct of the Study and Conduct of the Design Development.

A. Conduct of the Study

The researchers followed a series of steps in conducting the study as presented in the Gantt Chart in Appendix A. The first that has been undertaken is conducting a group meeting and identifying the whether the research project would be a training kit for ECE students or a robot for agricultural applications. The researchers consulted with their adviser first regarding the selection of research project. Then the topic/title presented, along with its concept paper, was subjected for evaluation during the title defense. After evaluation by the research instructor and panelists, the topic of Smart LoBo Kit: Hands-on IoT-based Logic Board Training Kit for ECE Students at University of Rizal System-Morong Campus was approved.

After the title defense, the researchers started writing the Chapter 1 and Chapter 2. The researchers consulted with their research adviser regarding the first two chapters. The comments and suggestions by the research adviser were noted for the revision of Chapter 1 and Chapter 2. Furthermore, more group meetings were conducted to take action what are the materials needed for the research project. The researchers also consulted with people who are knowledgeable to the topic interest to provide some helpful suggestions and

recommendations.

The next actions taken were the prototype development, testing, and evaluation. Data are collected, results were analyzed for the formulation of Chapter 3 which includes project design and computations, design results, and analysis and interpretation of result, and Chapter 4 which consists of summary of design results, conclusion, and recommendations.

The researchers consulted with their research adviser for the finalization and approval of the manuscript before it was submitted for the final defense. The researchers defended their thesis, showcasing the developed Logic Board Training Kit for ECE Students at the University of Rizal System-Morong. Finally, after addressing all the concerns of the adviser and panel members through manuscript revision, the settlement for the hardbound publication was organized and then the copies of the final manuscript were submitted.

B. Conduct of the Design Development

The researchers conducted this study entitled Smart LoBo Kit: Hands-on IoT-based Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus with the aim to develop an interactive IoT-integrated training board for logic circuits. This research provides a laboratory equipment for ECE students and Faculty members, specifically those who are taking and teaching the course of Digital Electronics 1: Logic Circuits and Switching Theory.

Firstly, the researchers analyzed the research topic by identifying the needs of the target population, gathering information from various sources including related studies on logic circuit design, articles, and online resources, and

interviews with faculty members and students who are potential users. In the planning stage, the researchers determined the technical and functional requirements of the system. The researchers created a block diagrams and schematic diagrams, as well as system flowcharts to demonstrate the system operations and interaction of components, outlining a clear set of goals for both the hardware and software components.

The next step involved the careful selection of appropriate hardware components. The researchers selected the ESP32 and ESP8266 microcontrollers to handle the system's processing needs. Various Integrated Circuits (ICs), consisting of different types of logic gates were used to construct logic circuits in this study. On the software side C++ was chosen as the primary programming language to control the microcontroller and manage system operations. The researchers used Microsoft Visual Studio Code as the Integrated Development Environment (IDE) for writing and debugging the code. Additionally, HTML and CSS were used to design the web application interface for the IoT system, allowing real-time update of problem sets. This combination of hardware and software was designed to enable accurate identification of the output from the logic circuits.

During the whole development process, different stages of testing were carried out to ensure the reliability and functionality of the developed device. This aspect of testing was characterized by continuous debugging and troubleshooting, specifically with the accuracy of the logic circuit output identification. Errors and inconsistencies in the program of the initial prototype were discovered, which resulted in several iterations of recoding and hardware adjustments to meet the

desired performance of the device. The level of acceptability of the device was evaluated through a survey questionnaire to gather feedback from target users, as well as experts in digital electronics. This feedback played an important role in identifying areas for improvement and further enhancing the system's design. Consequent to the feedback, adjustments were made so that Smart LoBo Training Kit met the practical needs of its users.

Project Design Model

In designing the Logic Board Training Kit, scientific and mathematical models were applied to ensure precise evaluation of the device's performance. A key model used in this study is the confusion matrix, which assesses the accuracy of the device in identifying logic circuit outputs.

The confusion matrix uses values such as true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) to evaluate how well the device's predictions match the actual logic outputs. Accuracy represents the proportion of correct predictions made by the device.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

The logic output generated by the Smart LoBo is compared to the

corresponding truth table produced by Logisim to assess its accuracy. Logisim, a logic circuit simulation tool, will generate the expected outputs for various logic circuits. The outputs from the Smart LoBo were tested against these expected results to determine how accurately the device replicates the correct logic behavior. By comparing the Smart LoBo's results with the Logisim-generated truth table, the accuracy of the device in identifying the correct logic circuit output can be calculated. This comparison is a critical step in evaluating the Smart LoBo's performance as a reliable training tool for logic circuit learning.

Statistical Treatment

This study employed a quantitative research approach to evaluate the performance of the developed Logic Board Training Kit. The confusion matrix is used to assess the accuracy of the device in identifying the logic outputs of the logic circuits. Additionally, the weighted mean was calculated to analyze survey responses. This method provided a clearer understanding of how well the training kit meets the expectations and requirements of its target users, as different Likert scale ratings were assigned varying weights. The analysis aided in refining the Logic Board Training Kit and ensure it adheres to necessary standards while offering valuable insights into its practical application as a learning tool. The formula for calculating weighted mean is presented in Appendix E.

Chapter 3

PRESENTATION, ANALYSIS, AND DESIGN RESULT

This chapter presents the result of design and development of the system, the presentation, analysis and interpretation of the result of its evaluation.

Design and Development of Smart Lobo Training Kit: Hands-On IoT-Based Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus

The development of Smart Lobo Training Kit: Hands-On IoT-Based Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus aims on enhancing student learning in digital electronics specifically on logic circuits. The core function of the Smart LoBo Training Kit is to replicate the features of a logic simulator software in a practical, hands-on device, providing students with an experience that bridges theoretical concepts and real-world applications. This approach allows learners to see the direct impact of logic circuit designs beyond software simulations, making concepts more tangible and comprehensible.

The device utilizes ESP32 and ESP8266 microcontrollers, enabling users to switch between Training Mode and IoT Mode. To better understand the components, refer to Appendix H. In Training Mode, the device provides a series of pre-determined questions designed to challenge users across three levels of difficulty: easy, average, and difficult. Each difficulty level is accompanied by a timer, ensuring that users work within a designated time frame. When the timer runs out, an audible buzzer signals the end of the allotted time. The corresponding question and timer information are displayed on the LCD screen, providing users with immediate visual feedback, as shown in Figure 5.

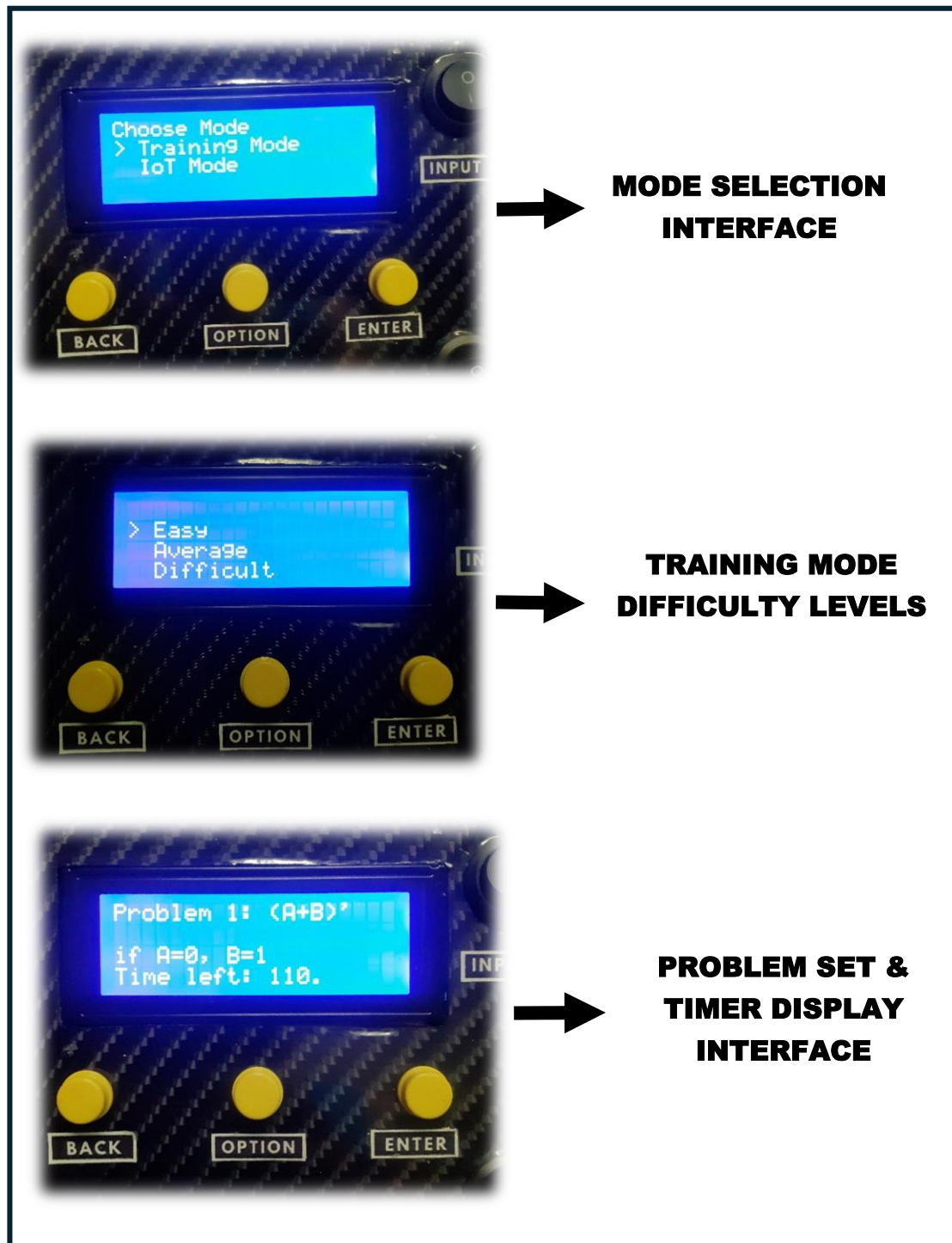


Figure 5

LCD Display Interface for Smart LoBo Training Kit: Hands-On IoT-Based Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus

As for the objectives 1.1, the IoT Mode enables wireless connectivity with Wi-Fi-enabled devices, allowing users to access a web server for remote configuration, including creating custom questions and adjusting timers. The web interface for IoT Mode is shown in Figure 6.

The figure displays three distinct web interface screens for the Smart LoBo Training Kit in IoT Mode, all featuring a light blue background and black text.

- Top Left Screen:** Titled "WELCOME TO Smart LoBo Training Kit (IoT-MODE)". It includes a label "Set Number of Questions (1-10):" followed by a text input field and a green "Submit" button.
- Top Right Screen:** Titled "Smart LoBo Training Kit" with the subtitle "Enter Problems, Timers, and Answers". It contains three input sections: "Problem 1:" with a text field, "Timer (seconds):" with a text field, and "Answer:" with a text field. Each section is followed by a green "Submit" button.
- Bottom Center Screen:** Titled "Smart LoBo Training Kit" with the message "The questions have been loaded, please check the display for the sequence." and a green button labeled "Want to try again?".

Figure 6

Web-Based Interface of IoT Mode for Smart LoBo Training Kit: Hands-On IoT-Based Logic Board Training Kit

Additionally, Figure 7 showcases a 3D design of the logic board, emphasizing its portability and user-friendly accessible design. The 3D model gives a clear understanding of the system's construction and layout, serving as a bridge from design expectations to the final actual device. This visualization provides insight into the design considerations that facilitate hands-on learning.

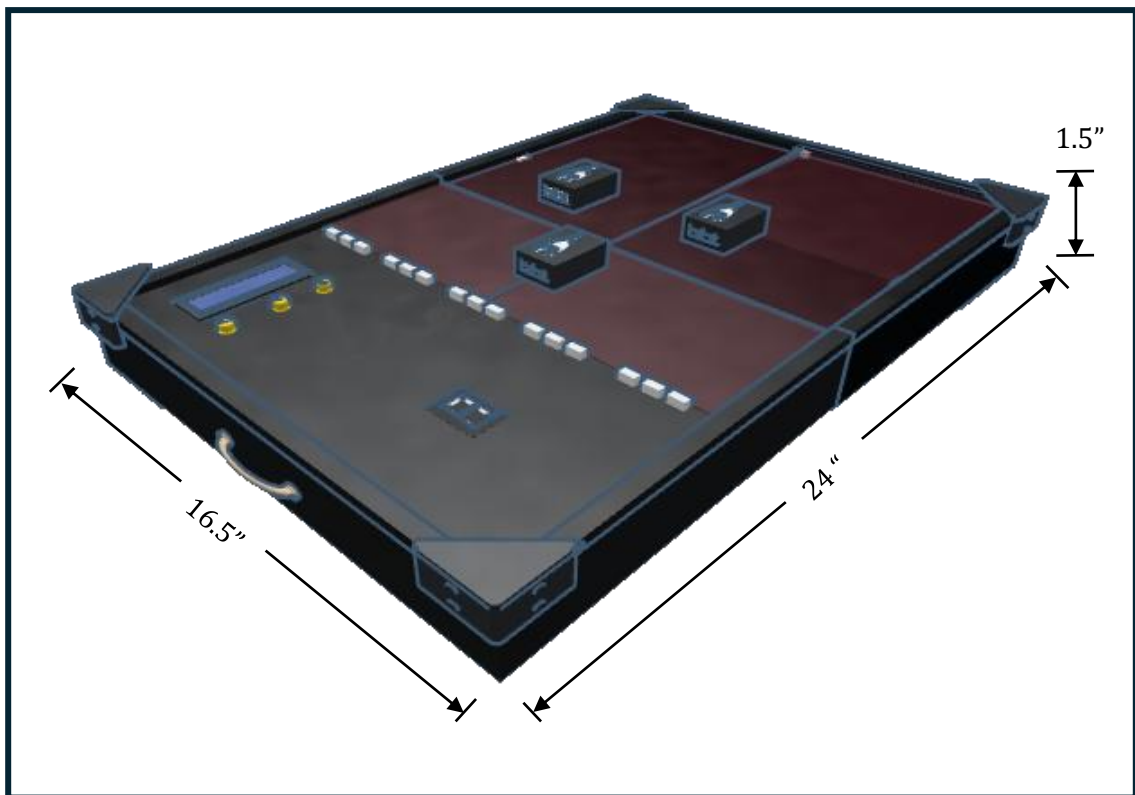
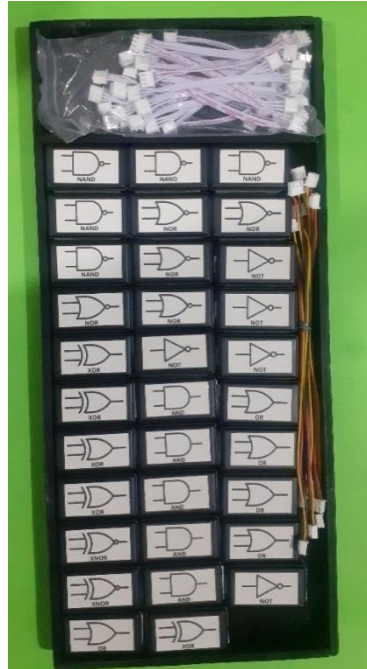


Figure 7

3D Model of the Smart LoBo Training Kit: Hands-On
IoT-Based Logic Board Training Kit

For the visual representation of the actual device, Figure 8 offers a pictorial representation of the Smart LoBo Training Kit. The device operates with a 9V, 2A power supply, featuring a barrel jack connector with negative polarity at the center and positive polarity on the outer side.



LOGIC BLOCKS



INTERNAL DESIGN

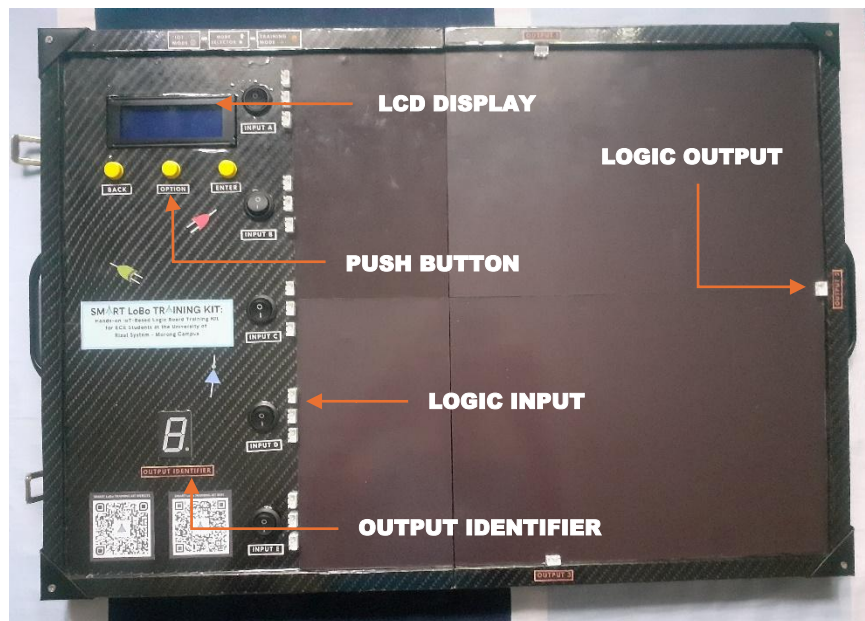


Figure 8

Pictorial of the Actual Device of Smart LoBo Training Kit: Hands-On IoT-Based Logic Board Training Kit

The power supply is converted to 5V using a voltage regulator, ensuring the safety of the components. The kit includes five inputs labeled from A to E, each equipped with three female JST connectors for linking the logic blocks that enables multiple connections. There are also three output ports, allowing users to choose one, accommodating situations where connection wires may be too short to reach other ports. The output identifier consists of a seven-segment display that clearly shows binary values (0s and 1s) to indicate the circuit's output state. The logic blocks are compact, with each containing integrated circuit that are marked clearly on their top and side with graphics and text naming them respectively so that learners can easily find and use them for their activities. Additionally, the logic blocks and board are equipped with magnetic sheets to keep the blocks secure and organized, avoiding disruptions similar to dislodged chess pieces.

The buttons are designed to be simple and intuitive, reminiscent of a classic game console. The mode switch is conveniently placed next to the power switch and clearly labeled for Training Mode and IoT Mode, ensuring users can easily identify and select the desired function.

For effective organization and troubleshooting, the electrical wiring is housed in a hinged cabinet-style panel board, allowing quick access to the internal components. This design choice makes maintenance and adjustments efficient, enhancing the overall usability of the training kit. Additionally, a user manual and learning modules are provided in the kit to help learners fully grasp the principles of logic circuits. The user manual and learning modules are presented in the Appendix J and K respectively.

Evaluate the System Performance of the Device in Identifying the Logic Output in Terms of Accuracy

The evaluation of the device's accuracy in identifying the logic output was conducted by comparing the Smart LoBo Training Kit's output to that of a logic simulator, specifically Logisim, as the simulator represents the ideal behavior for the logic circuits. Given that the logic training kit is designed to replicate the functions of a logic simulator in a physical format, it was essential to validate the accuracy of its outputs against the expected outputs from Logisim.

Table 1 illustrates the evaluation results using a confusion matrix, which measures the agreement between the outputs produced by the Smart LoBo Training Kit and those generated by Logisim.

Table 1

Evaluation of Accuracy of the Smart LoBo Training Kit: Hands-On IoT-Based Logic Board Training Kit

Difficulty Level	True Positive (1)	True Negative (0)	False Positive	False Negative	No. of Trials
Easy	5	14	1	0	20
Average	21	33	1	1	56
Difficult	18	20	1	1	40
Total	44	67	3	2	116

The data gathered in this evaluation was from Training Mode, where the logic board was tested against various levels of difficulty: Easy, Average, and Difficult. For each level, three different logic expressions were tested, each with distinct input values derived from the respective truth table for the logic gate.

Figure 9 shows the testing and comparison of the logic training kit and Logisim. The outputs of the Smart LoBo Training Kit were compared to the expected outputs generated by Logisim to assess the accuracy of the kit.

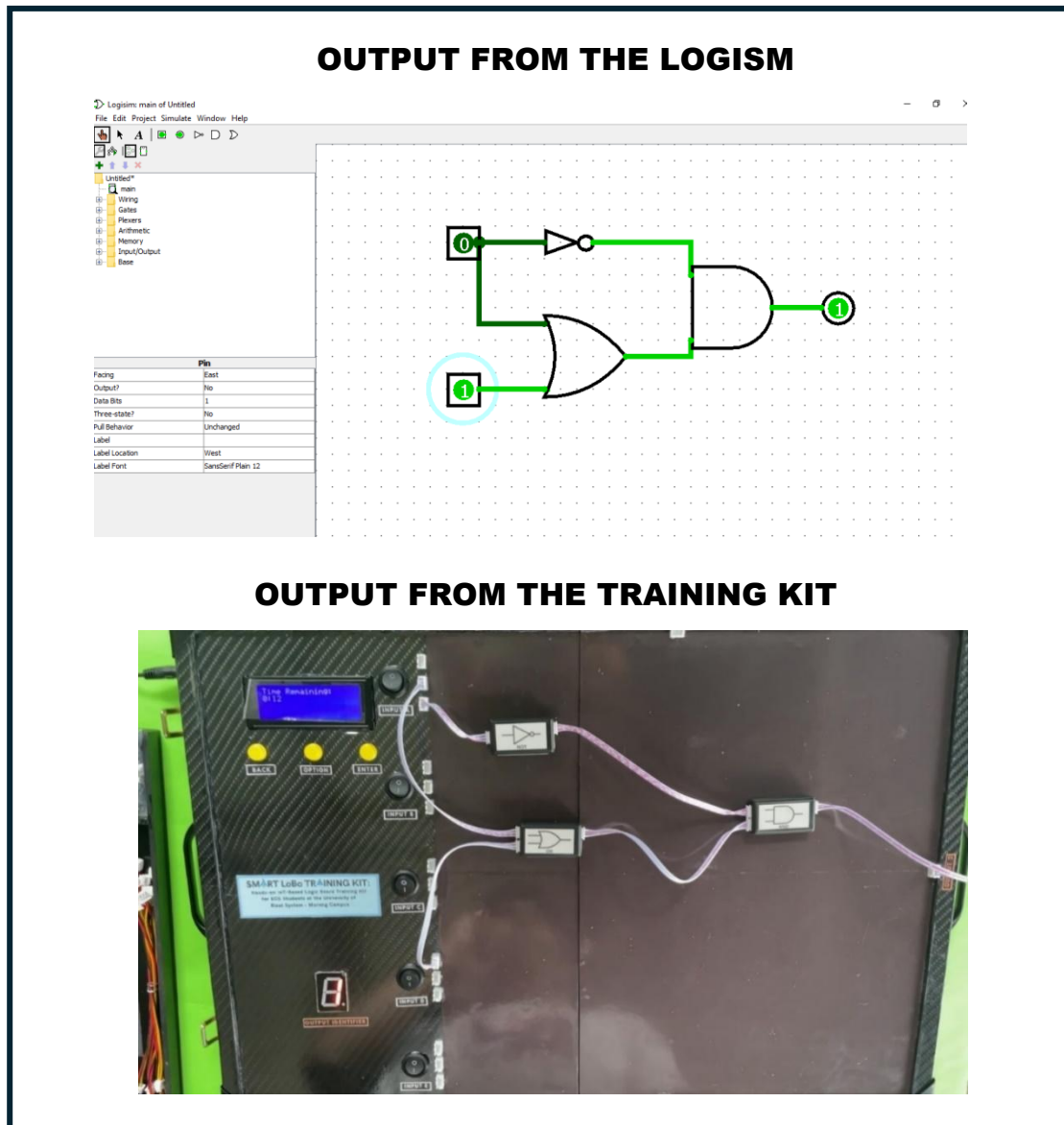


Figure 9

Sample Testing and Comparison of Smart LoBo Training Kit
to the Logisim Logic Circuit Simulator

This method ensured that the kit was tested with different inputs to simulate a wide range of logical operations. The results shows a 95.68 percent accuracy across all levels of difficulty, with minimal false positives or false negatives. This confirms that the Smart LoBo Training Kit correctly identified and displayed the logic output for each trial, exactly as expected. Since the device is designed to

mimic the behavior of a logic simulator like Logisim, the evaluation successfully demonstrates the high reliability of the kit for logic circuit. See Appendix F for the calculation of Logic kit output accuracy.

In this evaluation, a True Positive (TP) occurs when both the expected value from Logisim and the actual output from the Smart LoBo Training Kit are 1, while a True Negative (TN) occurs when both the expected and actual outputs are 0. A False Positive (FP) is recorded when the expected value is 0 but the actual output from the kit is 1, and a False Negative (FN) occurs when the expected value is 1 but the kit's output is 0. The data used to evaluate the accuracy using confusion matrix is shown in Appendix D.

Despite having high accuracy, it should be noted that potential issues such as loose connections, interference, or faulty components may affect performance in practical use. These issues were encountered during the accuracy testing of the device, which contributed to the discrepancies in the output.

The Level of Acceptability of Smart LoBo Kit: Hands-On IoT-Based Logic Board Training Kit

The researchers were able to gather the acceptability of the project according to functional suitability, performance efficiency, usability, reliability, and maintainability. It shows how acceptable the system is. The following tables below show the results of the computed level of acceptability of the goals of the study.

Table 2 presents the computed weighted mean, verbal interpretation, and average weighted mean of the level of acceptability of the Smart LoBo Training Kit: Hands-on Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus in terms of functional suitability.

Table 2

Computed Weighted Mean of the Smart LoBo Training Kit: Hands-on Logic Board Training Kit for ECE Students at the University of Rizal System - Morong Campus in Terms of Functional Suitability

Functional Suitability	Weighted Mean	Verbal Interpretation
1.1 The training kit provides a hands-on learning experience for logic circuits.	4.85	Very Much Acceptable
1.2 The activities and tasks provided by the kit align well with the learning outcomes.	4.78	Very Much Acceptable
1.3 The training kit supports a wide range of logic problems, aiding in comprehensive learning.	4.70	Very Much Acceptable
Average Weighted Mean	4.78	Very Much Acceptable

Table 2 shows the observations on the functional suitability of the Smart LoBo Training Kit. Item 1.1, evaluating the hands-on learning experience, scored highest with a mean of 4.85, interpreted as Very Much Acceptable. This is followed by item 1.2, which assesses activity alignment with learning outcomes, scoring 4.78, and item 1.3, evaluating support for diverse logic problems, scoring 4.70—all in the Very Much Acceptable range. An overall average weighted mean of 4.78 indicates strong acceptance, suggesting the kit effectively meets users' functional suitability needs.

These findings demonstrate respondents' high regard for the kit's functionality, highlighting it as a reliable educational tool for practical learning of the logic circuits. Supporting this, Farhan et al. (2023) emphasize the importance of educational tools that link theory and practice. The Smart LoBo Kit aligns with these principles by providing an engaging hands-on experience, supporting a range of logic problems, and helping students master essential concepts.

Table 3 presents the computed weighted mean, verbal interpretation, and average weighted mean of the level of acceptability of the Smart LoBo Training Kit: Hands-on Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus in terms of performance efficiency.

Table 3

Computed Weighted Mean of the Smart LoBo Training Kit: Hands-on Logic Board Training Kit for ECE Students at the University of Rizal System - Morong Campus in Terms of Performance Efficiency

Performance Efficiency	Weighted Mean	Verbal Interpretation
2.1 The training kit for executing logic circuits can quickly respond to user inputs during hands-on exercises.	4.72	Very Much Acceptable
2.2 The training kit provides smooth performance without interruptions during extended use.	4.67	Very Much Acceptable
2.3 The processing capacity of the training kit is sufficient for advanced logic circuit exercises.	4.73	Very Much Acceptable
Average Weighted Mean	4.71	Very Much Acceptable

Table 3 shows the observations of the respondents regarding the performance efficiency of the Smart LoBo Training Kit. Item 2.3, which evaluates the processing capacity of the training kit for advanced logic circuit exercises, received the highest mean of 4.73, interpreted as Very Much Acceptable, followed by item 2.1 with a mean of 4.72, also interpreted as Very Much Acceptable for its responsiveness to user inputs during hands-on exercises. Item 3.2, which assesses the kit's smooth performance without interruptions during extended use, received the lowest mean of 4.67, yet still falls within the Very Much Acceptable range. According to the table, respondents agreed that the developed system is

highly acceptable in terms of performance efficiency, achieving an overall weighted mean of 4.71.

This result implies that the Smart LoBo Training Kit effectively meets the needs, desires, and expectations of the users regarding performance efficiency, performing reliably under normal operating conditions. The system demonstrates quick response times, smooth uninterrupted performance during extended use, and sufficient processing power for advanced logic circuit exercises. Overall, the Smart LoBo Training Kit was found to be highly effective and satisfactory in meeting the performance efficiency requirements of its users.

To further validated the performance efficiency of the developed Smart LoBo Training Kit, the use of CD4511 CMOS decoders, known for their fast processing speed, plays a key role in the output identifier. According to Sehgal (2018), dynamic logic circuits, like those using CMOS technology, are essential for achieving high-speed and efficient operation in digital systems. The integration of the CD4511 decoder ensures that the Smart LoBo Training Kit delivers accurate and quick responses, supporting its overall effectiveness in assisting students with their logic circuit exercises.

Table 4 provides a comprehensive analysis of the usability of the Smart LoBo Training Kit: Hands-on Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus. This table presents the computed weighted mean for each usability metric, offering valuable insights into specific aspects of the kit's performance in practical, educational settings. It also includes verbal interpretations for each metric, which provide qualitative context to

complement the quantitative data. Furthermore, the table calculates the average weighted mean for usability, providing an overall assessment of the training kit's effectiveness in terms of user-friendliness and ease of use.

Table 4

Computed Weighted Mean of the Smart LoBo Training Kit: Hands-on Logic Board Training Kit for ECE Students at the University of Rizal System - Morong Campus in Terms of Usability

Usability	Weighted Mean	Verbal Interpretation
3.1 The kit is easy to set up and use for new users.	4.65	Very Much Acceptable
3.2 The interface of the training kit is intuitive and user-friendly.	4.73	Very Much Acceptable
3.3 The design of the training kit allows for comfortable and efficient handling during activities.	4.77	Very Much Acceptable
Average Weighted Mean	4.72	Very Much Acceptable

Table 4 shows the observations on the usability of the Smart LoBo Training Kit. Item 3.3, assessing the kit's comfortable and efficient design, received the highest mean of 4.77, rated as Very Much Acceptable. Item 3.2, evaluating the user-friendly interface, scored 4.73, while item 3.1, assessing ease of setup for new users, scored 4.65—all within the Very Much Acceptable range. The overall average mean of 4.72 indicates high usability, meeting users' expectations.

These findings suggest that the Smart LoBo Kit's intuitive design, ease of setup, and comfortable handling enhance the user experience, making it accessible even for new users. This aligns with Mohammad et al. (2018), whose study on the E-Logic Trainer Kit shows that user-friendly educational tools improve learning through practical simulations. The Smart LoBo Kit, similarly, demonstrates

success in enhancing learning outcomes through its intuitive and interactive design.

Table 5 presents the computed weighted mean on the level of acceptability of the Smart LoBo Training Kit: Hands-on Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus in terms of reliability.

Table 5

Computed Weighted Mean of the Smart LoBo Training Kit: Hands-on Logic Board Training Kit for ECE Students at the University of Rizal System - Morong Campus in Terms of Reliability

Reliability	Weighted Mean	Verbal Interpretation
The training kit accurately identifies the output of the logic circuits.	4.80	Very Much Acceptable
The kit performs reliably without frequent issues or malfunctions.	4.37	Very Much Acceptable
The kit performs consistently over repeated uses without requiring frequent maintenance.	4.70	Very Much Acceptable
Average Weighted Mean	4.62	Very Much Acceptable

Table 5 shows the observations of the respondents regarding the reliability of the Smart LoBo Training Kit. Item 4.1, which evaluates the accuracy of the kit in identifying the output of logic circuits, received the highest mean of 4.80, interpreted as Very Much Acceptable, followed by item 4.3, which assesses the kit's consistent performance over repeated uses without requiring frequent maintenance, with a mean of 4.70. Item 4.2, which evaluates the kit's reliability in performing without frequent issues or malfunctions, received a mean of 4.37, also falling within the Very Much Acceptable range. According to the table, the respondents agreed that the training kit is very much acceptable in terms of

reliability, with an overall weighted mean of 4.62.

These results imply that the Smart LoBo Training Kit is highly reliable and meets the users' expectations. The kit consistently provides accurate outputs for logic circuits and performs without frequent malfunctions, ensuring that students can rely on it for repeated use in their learning activities. This reliability not only enhances the learning experience but also reinforces the kit's role as an essential tool in supporting consistent and effective educational outcomes, which contributes to its overall acceptability.

To further reinforce the reliability of the developed system, the findings of Bocheng et al. (2014) highlight the importance of ensuring that digital logic teaching tools function smoothly with minimal issues, providing consistent and accurate results, particularly when used in educational environments. In alignment with this, the Smart LoBo Training Kit has been tested by students and has proven to deliver dependable performance, operating with minimal disruptions and consistently providing accurate results. This reliability aligns with the expectations of users, demonstrating its effectiveness for repeated use in educational settings. Thus, the Smart LoBo Training Kit meets the criteria for a robust and reliable educational tool, ensuring students can benefit from a stable and effective learning experience.

Table 6 presents the computed weighted mean on the level of acceptability of the Smart LoBo Training Kit: Hands-on Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus in terms of maintainability.

Table 6

Computed Weighted Mean of the Smart LoBo Training Kit: Hands-on Logic Board Training Kit for ECE Students at the University of Rizal System - Morong Campus in Terms of Maintainability

Maintainability	Weighted Mean	Verbal Interpretation
The training kit's design allows easy maintenance and troubleshooting.	4.77	Very Much Acceptable
The training kit's components can be replaced or repaired with minimal difficulty.	4.75	Very Much Acceptable
The kit is designed to minimize wear and tear during regular use.	4.68	Very Much Acceptable
Average Weighted Mean	4.73	Very Much Acceptable

Table 6 shows the observations of the respondents regarding the maintainability of the Smart LoBo Training Kit. Item 2.1, which assesses the design of the training kit for easy maintenance and troubleshooting, received the highest mean of 4.77, interpreted as Very Much Acceptable, followed by item 2.2, which evaluates the ease of replacing or repairing the kit's components, with a mean of 4.75, also interpreted as Very Much Acceptable. Item 2.3, which examines how well the kit is designed to minimize wear and tear during regular use, received a mean of 4.68, still within the Very Much Acceptable range. According to the table, the respondents agreed that the developed training kit is very much acceptable in terms of maintainability, with an average weighted mean of 4.73.

To validate the result further for the maintainability, Ajao (2017) highlights the importance of designing affordable digital logic training modules that simplify maintenance and promote hands-on experimentation, ensuring ease of use and efficient troubleshooting for students during laboratory exercises. The Smart LoBo

Training Kit follows this approach with its chessboard-style design making it familiar and intuitive to students, incorporating cost-effective components that make maintenance straightforward and accessible, ensuring long-term reliability and usability in educational settings.

Table 7 presents the summary of the level of acceptability of the Smart LoBo Training Kit: Hands-on Logic Board Training Kit for ECE Students at the University of Rizal System-Morong Campus. It is very much acceptable in terms of functional suitability, performance efficiency, usability, reliability, and maintainability.

Table 7

Composite Table on the Level of Acceptability of the Smart LoBo Training Kit:
Hands-on Logic Board Training Kit for ECE Students at the University
of Rizal System - Morong Campus

Variables	Mean	Verbal Interpretation
Functional Suitability	4.78	Very Much Acceptable
Performance Efficiency	4.71	Very Much Acceptable
Usability	4.72	Very Much Acceptable
Reliability	4.62	Very Much Acceptable
Maintainability	4.73	Very Much Acceptable
Overall Average Weighted Mean	4.71	Very Much Acceptable

As observed in Table 7, the functional suitability of the Smart LoBo Training Kit ranked first, achieving an average weighted mean of 4.78. Second in the ranking is maintainability, having achieved a weighted average of 4.73. The third in ranking is usability, which has an average weighted mean of 4.72. Performance efficiency follows, with an average weighted mean of 4.71. Lastly, reliability

received a mean of 4.62. Overall, the result displays a weighted average of 4.71 for all variables. This weighted average is interpreted as very much acceptable.

It simply implies that the developed Smart LoBo Training Kit: Hands-on Logic Board Training Kit for ECE Students is very much accepted and could state the facts in terms of functional suitability, performance efficiency, usability, reliability, and maintainability.

Chapter 4

SUMMARY OF DESIGN RESULTS, CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the summary of design results, conclusions, and recommendations of the study.

Summary of Design Results

The summary of design results stated as follows were established based on the evaluation made by the respondents.

1. The Smart LoBo Training Kit: Hands-on IoT-based Logic Board for ECE students at the University of Rizal System-Morong Campus, was successfully designed and developed with the intended purpose of enhancing learning in digital electronics, particularly logic circuits. The device operates with ESP32 and ESP8266 microcontrollers and offers two modes: Training Mode, for practicing logic circuit problems with varied difficulty levels on a timed basis, and IoT Mode, which allows logic circuit problems customization via a web-based application. It includes clearly labeled input and output ports, a seven-segment display for binary output, and a compact, portable design, making it user-friendly and accessible. A hinged panel board provides easy access to internal wiring, simplifying maintenance. Additionally, a comprehensive user manual and training module were created to guide students in using the kit effectively, supporting instructional content on logic gates and Boolean algebra. Altogether, the Smart LoBo Training Kit fulfills its purpose of providing a hands-on, structured learning experience that bridges theory and practical application for students.

2. The Smart LoBo Training Kit's accuracy in identifying logic outputs was evaluated by comparing its outputs to those of the Logisim logic simulator, which represents the ideal behavior of logic circuits. Using a confusion matrix, the evaluation measured the alignment between outputs from the training kit and Logisim across Easy, Average, and Difficult difficulty levels. In Training Mode, the kit was tested with, a total of nine distinct logic expressions (three per difficulty level) and input values derived from truth tables, achieving an accuracy rate of 95.68% across all levels, with minimal false positives and false negatives. This high accuracy demonstrates the kit's reliability in replicating a logic simulator's behavior for educational purposes.
3. A survey was conducted with 60 participants, comprising 13 experts and 47 students, to determine the level of acceptability of the Smart LoBo Training Kit: Hands-on IoT-based Logic Board for ECE students at the University of Rizal System - Morong Campus. The findings revealed a strong level of approval across various criteria. Functional suitability received an average weighted mean of 4.78, which was verbally interpreted as very much acceptable. In terms of performance efficiency, the system achieved a weighted mean of 4.71, also interpreted as very much acceptable. For maintainability, the device received a mean of 4.73, interpreted as very much acceptable. Usability was rated with a weighted mean of 4.72, verbally interpreted as very much acceptable. Lastly, the reliability of the device scored a mean of 4.62, interpreted as very much acceptable. The overall average weighted mean of 4.71 further indicated that the training kit was very much acceptable across all variables.

Conclusions

The following conclusions were drawn based on the aforementioned findings of the developed Smart LoBo Training Kit: Hands-on IoT-based Logic Board for ECE students at the University of Rizal System-Morong Campus:

1. The Smart LoBo Training Kit effectively meets its design objectives, providing an accessible, user-friendly, and practical tool for learning digital electronics, particularly logic circuits. By integrating ESP32 and ESP8266 microcontrollers, the device offers versatile functionality with both Training and IoT Modes, allowing students to engage with customized logic circuit problems. The user manual and training module provided in the kit effectively guide the students in using the kit and in experiencing hands-on application that aligns with theoretical concepts.
2. The accuracy evaluation showed that the Smart LoBo Training Kit closely replicates the expected behavior of a logic simulator software application, achieving a 95.68% accuracy rate compared to Logisim. This high degree of reliability in logic output verification confirms the kit's effectiveness in mirroring an ideal logic circuit environment, making it a dependable tool for education.
3. The survey conducted with 60 participants, including 13 experts and 47 students, demonstrated a notable level of satisfaction across multiple metrics. The device scored impressively high on functional suitability, performance efficiency, usability, reliability, and maintainability, with ratings ranging from 4.62 to 4.78 on the weighted mean scale. Overall, these results underscore the device's acceptance garnering positive feedback from students and experts.

Recommendations

Based on the results of the study, the researchers arrived on the following recommendations:

1. The researchers propose the use of detachable/rechargeable battery for maximizing the portability.
2. The researchers recommend to use single microcontroller to minimize the circuitry and to make it less complex.
3. Replacing the LCD to TFT touch screen display will improve the user interface and reduce the use of push buttons as well as the limitation of text and lines in 20x4 LCD.
4. Future researchers are encouraged to continuously improve the program of the device to enhance the efficiency of future ECE student's analytical abilities to logic circuit.

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Gates. Department of Education - Bureau of Learning Resources (DepEd BLR)

APPENDICES

Appendix B

Appointment Letters



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Province of Rizal
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College of Engineering - URS Morong

APPOINTMENT

August 30, 2024

ENGR. JENADEL DL. ANTIPOLLO

Faculty
This College

Dear Madame;

In consideration of your qualifications in the field of research, the College Research Council represented by the undersigned, upon the recommendations of the Research Professor has approved your appointment as **Research Adviser**.

The following are the responsibilities of the Research Adviser:

1. Supervises and monitors the research work of the advisees upon accepting the assignment as an adviser;
2. Sees that thesis meets the requirements of research before endorsing to the members of the panel and research instructor for consultation;
3. Certifies to the readiness of the advisees to render final defense.
4. Requires the advisees to give a copy of the manuscript to the panel members one week before the scheduled oral defense
5. Works closely with the advisees and monitors the revision of the manuscript considering the suggestions of the panel and certifies the correctness of the same before referring to the members of the panel and research instructors;
6. Coordinates with the College Research Coordinator on matters related to research activities; and
7. Undertakes the function of the recorder who records proceedings of the defense and supplies the advisees with the summarized recommendations / suggestions of the panel.

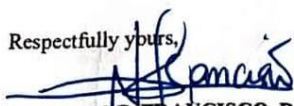
The students who shall be under your supervision include:

<i>Names</i>	<i>Course</i>
Ashanty Madeline DC. Cruz	BS ECE
Brian Ezekiel D. Batalon	BS ECE
Tristan Manuel T. Sto. Tomas	BS ECE
Mark Jayson F. Villacura	BS ECE

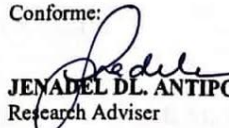
The research project proposal is entitled **SMART LoBo TRAINING KIT: HANDS-ON IoT-BASED LOGIC BOARD TRAINING KIT FOR ECE STUDENTS AT THE UNIVERSITY OF RIZAL SYSTEM - MORONG CAMPUS.**

Thank you for your usual support to the research program of the College.

Respectfully yours,


NORMAN C. FRANCISCO, REE
College Research Coordinator

Conforme:


JENADEL DL. ANTIPOLLO, ECE
Research Adviser

Nurturing Tomorrow's Noblest

URS Angono
URS Antipolo
URS Binangonan

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Tel. 8539-9935 to 37

URS Calinta
URS Cardona
URS Morong

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Tel. 8539-9940 to 41
Tel. 8539-9950 to 56

URS Piliila
URS Rodriguez
URS Taytay

Tel. 8539-9942 to 44
Tel. 8539-9945 to 47
Tel. 8539-9948 to 49



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College of Engineering - URS Morong

APPOINTMENT

August 30, 2024

ENGR. RIO CAMILLE M. PEDROCILLO

Faculty
This College

Dear Madame;

In consideration of your qualifications in the field of research, the College Research Council represented by the undersigned, upon the recommendations of the Research Professor has approved your appointment as **Panel Chairperson**.

The following are the responsibilities of the Panel Chairperson:

1. Acts as presiding officer;
2. Decides on the procedure of the defense;
3. Settles disagreements among the panel members and/or comments and suggestions;
4. Solicits the decision of the panel, and
5. Consolidates and announces the rating.

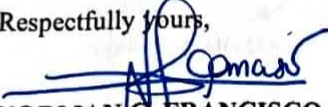
The students who shall be under your scrutiny include:

<i>Names</i>	<i>Course</i>
Ashanty Madeline DC. Cruz	BS ECE
Brian Ezekiel D. Batalon	BS ECE
Tristan Manuel T. Sto. Tomas	BS ECE
Mark Jayson F. Villacura	BS ECE

The research project proposal is entitled **SMART LoBo TRAINING KIT: HANDS-ON IoT-BASED LOGIC BOARD TRAINING KIT FOR ECE STUDENTS AT THE UNIVERSITY OF RIZAL SYSTEM - MORONG CAMPUS**.

Thank you for your usual support to the research program of the College.

Respectfully yours,


NORMAN C. FRANCISCO, REE
College Research Coordinator

Conforme:


RIO CAMILLE M. PEDROCILLO, ECE
Panel Chairperson

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College of Engineering - URS Morong

APPOINTMENT

August 30, 2024

ENGR. CRYSTALEENE JADE A. SANTOS

Faculty

This College

Dear Madame;

In consideration of your qualifications in the field of research, the College Research Council represented by the undersigned, upon the recommendations of the Research Professor has approved your appointment as **Research Expert**.

The following are the responsibilities of the Thesis Expert:

1. Provide subject matter expertise for the study;
2. Suggests appropriate statistical treatment for the study;
3. Guides the researchers in computation of data;
4. Guides the researchers in the correct analysis and interpretation of data.

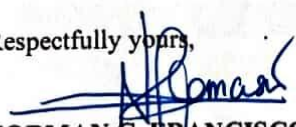
The students who shall be under your scrutiny include:

<i>Names</i>	<i>Course</i>
Ashanty Madeline DC. Cruz	BS ECE
Brian Ezekiel D. Batalon	BS ECE
Tristan Manuel T. Sto. Tomas	BS ECE
Mark Jayson F. Villacura	BS ECE

The research project proposal is entitled **SMART LoBo TRAINING KIT: HANDS-ON IoT-BASED LOGIC BOARD TRAINING KIT FOR ECE STUDENTS AT THE UNIVERSITY OF RIZAL SYSTEM - MORONG CAMPUS**.

Thank you for your usual support to the research program of the College.

Respectfully yours,


NORMAN C. FRANCISCO, REE
College Research Coordinator

Conforme:


CRYSTALEENE JADE A. SANTOS, CpE
Research Expert

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College of Engineering - URS Morong

APPOINTMENT

August 30, 2024

ENGR. ERICKSON T. MARCOS

Faculty
This College

Dear Sir;

In consideration of your qualifications in the field of research, the College Research Council represented by the undersigned, upon the recommendations of the Research Professor has approved your appointment as **Critic Reader**.

The following are the responsibilities of the Critic Reader:

1. Reviews all the grammatical aspects;
2. Certifies that the manuscript is already free of grammatical errors and is ready for submission to the office of the Dean of the College;

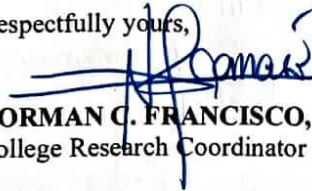
The students who shall be under your scrutiny include:

<i>Names</i>	<i>Course</i>
Ashanty Madeline DC. Cruz	BS ECE
Brian Ezekiel D. Batalon	BS ECE
Tristan Manuel T. Sto. Tomas	BS ECE
Mark Jayson F. Villacura	BS ECE

The research project proposal is entitled **SMART LoBo TRAINING KIT: HANDS-ON IoT-BASED LOGIC BOARD TRAINING KIT FOR ECE STUDENTS AT THE UNIVERSITY OF RIZAL SYSTEM - MORONG CAMPUS**.

Thank you for your usual support to the research program of the College.

Respectfully yours,


NORMAN C. FRANCISCO, REE
College Research Coordinator

Conforme:


ERICKSON T. MARCOS, MSc
Critic Reader

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URS Antipolo
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College of Engineering - URS Morong

**DESIGNATION OF METHODS OF RESEARCH PROJECT ADVISORY/ ORAL
EXAMINATION COMMITTEE**

Date: September 12, 2024

TO: Engr. Rio Camille M. Pedrocillo
Engr. Crystaleene Jade A. Santos
Engr. Erickson T. Marcos

We have the honor to inform you that you are hereby requested as Adviser, Chairperson, Critic Reader and Expert for Ashanty Madeline DC. Cruz, Brian Ezekiel D. Batalon, Tristan Manuel T. Sto. Tomas, and Mark Jayson F. Villacura students of **BS Electronics Engineering** as follows:

Panel Chairman: Rio Camille M. Pedrocillo, ECE
Expert: Crystaleene Jade A. Santos, CpE
Critic Reader: Erickson T. Marcos, MSc

CONFORME

[Signature]
[Signature]
[Signature]

The approved project title is SMART LOBO TRAINING KIT: HANDS-ON IOT-BASED LOGIC BOARD TRAINING KIT FOR ECE STUDENTS AT THE UNIVERSITY OF RIZAL SYSTEM – MORONG CAMPUS, an outline of which is herewith attached for your perusal, evaluation and approval.

As such, you are humbly asked to advise and guide the advisee during the operation conduct of the project to ensure high scholastic standards of the research.

Thank you very much.

Very truly yours,

[Signature]
Jenadel DL. Antipolo, ECE
Research Adviser, Research Instructor,
Electronics Engineering

Noted by:

[Signature]
Michael L. Pascua, MSCM
Dean, College of Engineering

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Appendix C

Communication Letters and Forms



Republic of the Philippines
UNIVERSITY OF RIZAL SYSTEM
MORONG, RIZAL

COLLEGE OF ENGINEERING
Bachelor of Science in Electronics Engineering



**RESEARCHER-MADE QUESTIONNAIRE CHECKLIST FOR THE RESEARCH
PROJECT SMART LoBo TRAINING KIT: HANDS-ON IOT-BASED
LOGIC BOARD TRAINING KIT FOR ECE STUDENTS AT THE
UNIVERSITY OF RIZAL SYSTEM - MORONG CAMPUS**

Part I. Personal Data (Optional)

Name: _____ Date: _____
(Pangalan) (Petsa)

☐

Expert

☐

Student

Part II. Level of Acceptability of Smart LoBo Training Kit: Hands-on IoT-Based Logic Board Training Kit for ECE Students at the University of Rizal System - Morong Campus

Direction: Using the scale below, rate the Smart LoBo Training Kit: Hands-on IoT-Based Logic Board Training Kit for ECE Students according to the criteria by ensuring that the relevant column is checked.

(Direksyon: Gamit ang sukat sa ibaba, i-rate ang Smart LoBo Training Kit: Hands-on IoT-Based Logic Board Training Kit for ECE Students ayon sa pamantayan sa pamamagitan ng pagtiyak na ang nauugnay na column ay nalagyan ng tsek.)

Scales are as follows:

5 – Very Much Acceptable (*Sobrang katanggap-tanggap*)

4 – Much Acceptable (*Lubhang katanggap-tanggap*)

3 – Acceptable (*Katanggap-tanggap*)

2 – Slightly Acceptable (*Bahagyang katanggap-tanggap*)

1 – Not Acceptable (*Hindi katanggap-tanggap*)

1. Functional Suitability	5	4	3	2	1
1.1 The training kit provides a hands-on learning experience for logic circuits. <i>(Ang training kit ay nagbibigay ng hands-on na karanasan sa pag-aaral sa mga logic circuit.)</i>					
1.2 The activities and tasks provided by the kit align well with the learning outcomes. <i>(Ang mga aktibidad at gawain na ibinigay ng kit ay mahusay na umaayon sa mga layunin ng pagkatuto.)</i>					
1.3 The training kit supports a wide range of logic problems, aiding in comprehensive learning. <i>(Ang training kit ay sumusuporta sa malawak na hanay ng mga problems sa logic circuit, na tumutulong sa komprehensibong pagkatuto.)</i>					
2. Performance Efficiency					
2.1 The training kit for executing logic circuits can quickly respond to user inputs during hands-on exercises. <i>(Ang training kit para sa pag-execute ng mga logic circuit ay mabilis na rumeresponde sa mga input ng mga gumagamit sa oras ng hands-on exercises.)</i>					
2.2 The training kit provides smooth performance without interruptions during extended use. <i>(Ang training kit ay nagbibigay ng maayos na performance nang walang pagkaabala sa mahabang paggamit.)</i>					
2.3 The processing capacity of the training kit is sufficient for advanced logic circuit exercises. <i>(Ang kapasidad ng pagproseso ng training kit ay sapat para sa mga advanced na exercises sa logic circuit.)</i>					
3. Usability					
3.1 The kit is easy to set up and use for new users. <i>(Madaling i-set up at gamitin ang training kit para sa mga bagong gumagamit.)</i>					
3.2 The interface of the training kit is intuitive and user-friendly. <i>(Ang interface ng training kit ay madaling maunawaan.)</i>					
3.3 The design of the training kit allows for comfortable and efficient handling during activities. <i>(Ang disenyo ng training kit ay nagbibigay-daan sa komportable at mahusay na handling sa oras ng mga aktibidad.)</i>					
4. Reliability					
4.1 The training kit accurately identifies the output of the logic circuits.					

(Ang training kit ay tamang na-identify ang output ng mga logic circuit.)					
4.2 The kit performs reliably without frequent issues or malfunctions. (Ang training kit ay maaasahan at hindi madalas nagkakaroon ng mga isyu o pagkasira.)					
4.3 The kit performs consistently over repeated uses without requiring frequent maintenance. (Ang training kit ay patuloy na gumagana nang maayos sa paulit-ulit na paggamit nang hindi nangangailangan ng madalas na maintenance.)					
5. Maintainability					
5.1 The training kit's design allows easy maintenance and troubleshooting. (Ang disenyo ng training kit ay madaling mapanatili at maayos.)					
5.2 The training kit's components can be replaced or repaired with minimal difficulty. (Ang mga bahagi ng training kit ay maaaring palitan o ayusin nang may kaunting hirap.)					
5.3 The kit is designed to minimize wear and tear during regular use. (Ang training kit ay dinisenyo upang i-minimize ang pagkasira sa regular na paggamit.)					

Comments/Suggestions:



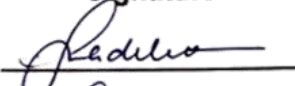
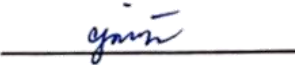
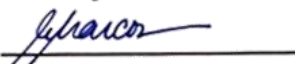
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MORONG, RIZAL



COLLEGE OF ENGINEERING
Bachelor of Science in Electronics Engineering

CERTIFICATE OF CONTENT VALIDATION

This is to certify that the following experts have validated the researchers' questionnaire checklist prepared and submitted by Ashanty Madeline DC. Cruz, Brian Ezekiel D. Batalon, Tristan Manuel T. Santo Tomas, and Mark Jayson F. Villacura, for their undergraduate research project entitled SMART LoBo TRAINING KIT: HANDS-ON IOT-BASED LOGIC BOARD TRAINING KIT FOR ECE STUDENTS AT THE UNIVERSITY OF RIZAL SYSTEM - MORONG CAMPUS as a partial requirement for the degree Bachelor of Science in Electronics Engineering (BSECE).

Name	Signature	Date
Engr. Jenadel DL. Antipolo		11/05/2024
Engr. Rio Camille M. Pedrocillo		11/06/2024
Engr. Crystaleene Jade A. Santos		11/06/2024
Engr. Erickson T. Marcos		11/07/2024



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UNIVERSITY OF RIZAL SYSTEM
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LETTER TO THE RESPONDENTS

November 4, 2024

Dear Respondent,

We, the researchers, are conducting the research study entitled SMART LoBo TRAINING KIT: HANDS-ON IOT-BASED LOGIC BOARD TRAINING KIT FOR ECE STUDENTS AT THE UNIVERSITY OF RIZAL SYSTEM - MORONG CAMPUS in partial fulfillment of the requirements for the degree of Bachelor of Science in Electronics Engineering (BSECE). In light of this, we would like to have you as one of our respondents in this research study. We will appreciate it if you could accomplish the attached questionnaire checklist promptly and honestly.

Rest assured that your response would be treated with utmost confidentiality. We are hoping for your favorable response. Thank you very much.

Respectfully yours,

Ashanty Madeline DC. Cruz

Tristan Manuel T. Santo Tomas

Brian Ezekiel D. Batalon

Mark Jayson F. Villacura

Noted:

JENADEL DL. ANTIPOLLO, ECE
Research Adviser



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**APPLICATION FOR UNDERGRADUATE METHODS OF
RESEARCH DEFENSE**

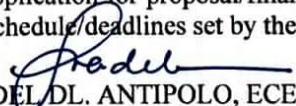
Student Information

Last Name	First Name	Middle Name	Student No.
CRUZ	ASHANTY MADELINE	DELA CRUZ	M2021-0601
BATALON	BRIAN EZEKIEL	DELA CRUZ	M2021-0617
STO. TOMAS	TRISTAN MANUEL	TEODORO	M2021-0595
VILLACURA	MARK JAYSON	FLORIN	M2021-0638

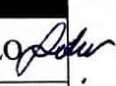
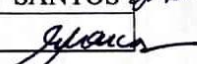
Academic Information

Program	BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING		
Year & Section	4 ECE-A	SY Semester	2024-2025

Thesis Information

Methods of Research Title	
SMART LoBo TRAINING KIT: HANDS-ON IOT-BASED LOGIC BOARD TRAINING KIT FOR ECE STUDENTS AT THE UNIVERSITY OF RIZAL SYSTEM - MORONG CAMPUS	
Defense applied for: ☐ Proposal Defense ☒ Final Defense	
Adviser's Certification I hereby certify that I have read the methods of research project manuscript by the students in connection with this application for proposal/final defense and classify the same as eligible for defense within the schedule/deadlines set by the university.  JENADEL DL. ANTIPOLO, ECE Signature Over Printed Name / Date	

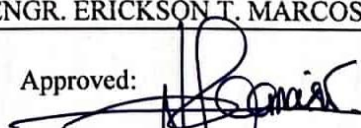
Schedule and Panel Members

Date/Time of Defense	PANEL MEMBERS	
November 18, 2024 / 1:00PM – 2:30PM	Chairperson:	ENGR. RIO CAMILLE M. PEDROCILLO 
Assigned Room	Expert:	ENGR. CRYSTALEENE JADE A. SANTOS 
Room 2C, SNT Building	Critic Reader:	ENGR. ERICKSON T. MARCOS 

Recommending Approval:


JENADEL DL. ANTIPOLO, ECE
Research Instructor

Approved:


NORMAN C. FRANCISCO, REE
College Research Coordinator

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URS Cardona
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Tel. 8539-9940 to 41
Tel. 8539-9950 to 56

URS Pililla
URS Rodriguez
URS Taytay

Tel. 8539-9942 to 44
Tel. 8539-9945 to 47
Tel. 8539-9948 to 49



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CERTIFICATE OF COMPLETION
(Undergraduate Project)

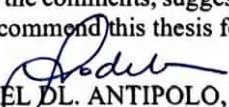
Student Information

Last Name	First Name	Middle Name	Student No.
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BATALON	BRIAN EZEKIEL	DELA CRUZ	M2021-0617
STO. TOMAS	TRISTAN MANUEL	TEODORO	M2021-0595
VILLACURA	MARK JAYSON	FLORIN	M2021-0638

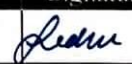
Academic Information

Program	BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING		
Year & Section	4 ECE-A	SY/Semester	2024-2025

Project Information

Project Title	
SMART LoBo TRAINING KIT: HANDS-ON IoT-BASED LOGIC BOARD TRAINING KIT FOR ECE STUDENTS AT THE UNIVERSITY OF RIZAL SYSTEM – MORONG CAMPUS	
Date/Time of Defense: November 19, 2024 (1:00-2:30pm)	Assigned Room: Room 2C, SNT Building
Adviser's Certification I hereby certify that I have read the revised thesis manuscript presented by the students as required by the members of the Oral Examination Committee. Furthermore, I certify that the revisions are in accordance with the comments, suggestions and recommendations during the final oral defense. As such, I recommend this thesis for approval.  JENADEL D. ANTIPOLLO, ECE Signature Over Printed Name / Date	

Panel Approval

Panel Members		Signature	Date
Chairperson	ENGR. RIO CAMILLE M. PEDROCILLO		12/03/24
Expert	ENGR. CRYSTALEENE JADE A. SANTOS		12/03/24
Critic Reader	ENGR. ERICKSON T. MARCOS		12/03/24

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URS Binangonan Tel. 8539-9935 to 37

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URS Rodriguez
URS Tawtaw

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Appendix D

Data Tables for Logic Board Output Accuracy Assessment

EASY MODE Q1: $A' \cdot (A + D) = X$							
A	D	Expected Output	Actual Output	TP	TN	FP	FN
1	1	0	0	0	1	0	0
0	1	1	1	1	0	0	0
1	0	0	0	0	1	0	0
0	0	1	1	1	0	0	0
Total				2	2		

EASY MODE Q2: $AB' + BC' = X$								
A	B	C	Expected Output	Actual Output	TP	TN	FP	FN
1	1	1	0	0		1		
0	1	1	0	0		1		
1	0	1	1	1	1			
0	0	1	0	0		1		
1	1	0	0	0		1		
0	1	0	0	1			1	
1	0	0	1	1	1			
0	0	0	0	0		1		
Total					2	5	1	

EASY MODE Q3: $(A'D) \cdot (AB') = X$								
A	B	D	Expected Output	Actual Output	TP	TN	FP	FN
1	1	1	0	0		1		
0	1	1	0	0		1		
1	0	1	0	0		1		
0	0	1	0	0		1		
1	1	0	0	0		1		
0	1	0	0	0		1		
1	0	0	1	1	1			
0	0	0	0	0		1		
Total					1	7		

AVERAGE Q1: $((BE) + (BE)') \cdot (C + E)' = X$								
B	E	C	Expected Output	Actual Output	TP	TN	FP	FN
1	1	1	0	0		1		
0	1	1	0	0		1		
1	0	1	0	0		1		
0	0	1	0	0		1		
1	1	0	0	1			1	
0	1	0	0	0		1		
1	0	0	1	0				1
0	0	0	1	1	1			
Total					1	5	1	1

AVERAGE Q2: $A'DE + (A' \cdot (AB)') + DE = X$									
A	B	C	D	Expected Output	Actual Output	TP	TN	FP	FN
1	1	1	1	0	0		1		
0	1	1	1	0	0		1		
1	0	1	1	0	0		1		
0	0	1	1	0	0		1		
1	1	0	1	0	0		1		
0	1	0	1	0	0		1		
1	0	0	1	0	0		1		
0	0	0	1	0	0		1		
1	1	1	0	0	0		1		
0	1	1	0	0	0		1		
1	0	1	0	1	1	1			
0	0	1	0	0	0		1		
1	1	0	0	1	1	1			
0	1	0	0	0	0		1		
1	0	0	0	1	1	1			
0	0	0	0	0	0		1		
Total						3	13		

AVERAGE Q5: $[(AE' \oplus B)] + [(B \oplus C'E) \cdot (B + (A'D)')]$											
A	E	B	C	D	Expected Output	Actual Output	TP	TN	FP	FN	
1	1	1	1	1	1	1	1	0	0	0	
0	1	1	1	1	0	0	0	1	0	0	
1	0	1	1	1	0	0	0	1	0	0	
0	0	1	1	1	0	0	0	1	0	0	
1	1	0	1	1	0	0	0	1	0	0	
0	1	0	1	1	1	1	1	0	0	0	
1	0	0	1	1	1	1	1	0	0	0	
0	0	0	1	1	1	1	1	0	0	0	
1	1	1	0	1	1	1	1	0	0	0	
0	1	1	0	1	0	0	0	1	0	0	
1	0	1	0	1	0	0	0	1	0	0	
0	0	1	0	1	0	0	0	1	0	0	
1	1	0	0	1	1	1	1	0	0	0	
0	1	0	0	1	1	1	1	0	0	0	
1	0	0	0	1	1	1	1	0	0	0	
0	0	0	0	1	1	1	1	0	0	0	
1	1	1	1	0	1	1	1	0	0	0	
0	1	1	1	0	0	0	0	1	0	0	
1	0	1	1	0	0	0	0	1	0	0	
0	0	1	1	0	0	0	0	1	0	0	
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1	0	0	1	0	1	1	1	0	0	0	
0	0	0	1	0	1	1	1	0	0	0	
1	1	1	0	0	1	1	1	0	0	0	
0	1	1	0	0	0	0	0	1	0	0	
1	0	1	0	0	0	0	0	1	0	0	
0	0	1	0	0	0	0	0	1	0	0	
1	1	0	0	0	0	0	0	1	0	0	
0	1	0	0	0	1	1	1	0	0	0	
1	0	0	0	0	1	1	1	0	0	0	
0	0	0	0	0	1	1	1	0	0	0	
Total							17	15	0	0	

DIFFICULT Q3: CD + AC' + C'D'								
C	D	A	Expected Output	Actual Output	TP	TN	FP	FN
1	1	1	1	1	1	0	0	0
0	1	1	0	0	0	1	0	0
1	0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0	0
1	1	0	1	1	1	0	0	0
0	1	0	0	0	0	1	0	0
1	0	0	0	0	0	1	0	0
0	0	0	0	0	0	1	0	0
Total					2	6	0	0

DIFFICULT Q4: AB + BD + C'									
A	B	C	D	Expected Output	Actual Output	TP	TN	FP	FN
1	1	1	1	1	1	1	0	0	0
0	1	1	1	1	1	1	0	0	0
1	0	1	1	0	1	0	0	1	0
0	0	1	1	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	1	0	0
1	1	0	1	1	1	1	0	0	0
1	0	0	1	1	1	1	0	0	0
0	0	0	1	1	1	1	0	0	0
1	1	0	0	1	1	1	0	0	0
0	1	0	0	1	1	1	0	0	0
1	0	0	0	1	1	1	0	0	0
0	0	0	0	1	1	0	1	0	0
1	1	1	0	1	1	1	0	0	0
1	0	1	0	0	0	0	1	0	0
0	1	0	1	1	1	1	0	0	0
1	1	1	1	1	1	1	0	0	0
Total						10	5	1	0

DIFFICULT Q5: C' + B'D'E'									
B	C	D	E	Expected Output	Actual Output	TP	TN	FP	FN
1	1	1	1	0	0	0	1	0	0
1	0	1	1	0	0	0	1	0	0
0	1	1	1	0	0	0	1	0	0
0	0	1	1	1	1	1	0	0	0
1	1	0	1	0	0	0	1	0	0
1	0	0	1	1	1	1	0	0	0
0	1	0	1	0	0	0	1	0	0
0	0	0	1	1	1	1	0	0	0
1	1	1	0	0	0	0	1	0	0
1	0	1	0	1	1	1	0	0	0
0	1	1	0	0	0	0	1	0	0
0	0	1	0	1	1	1	0	0	0
1	1	0	0	0	0	0	1	0	0
1	0	0	0	1	1	1	0	0	0
0	1	0	0	0	0	0	1	0	0
0	0	0	0	1	0	0	0	0	1
Total						6	9	0	1

Appendix E

Formula/Equation used in the Study

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

P = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

Weighted Mean:

$$\bar{X} = \frac{\sum x}{N}$$

Where:

$\bar{X}$ = Mean

Σ = sum of

x = a rows score in a set of scores

N = Total number of scores in a set

Appendix F

Calculation of Accuracy of the device

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{44 + 67}{44 + 67 + 3 + 2} = 0.9568 \times 100$$

$$Accuracy = 95.68\%$$

Where:

P = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

Appendix G

System's Code

```

esp32_practice_mode | Arduino IDE 2.2.0
File Edit Sketch Tools Help
ESP32-WROOM-0A Mod

esp32_practice_mode.ino
1 #include <Wire.h>
2 #include <LiquidCrystal_I2C.h>
3
4 #define BUTTON_SELECT 26
5 #define BUTTON_OPTION 25
6 #define BUTTON_BACK 33
7 #define BUZZER_PIN 32
8
9 LiquidCrystal_I2C lcd(0x27, 20, 4);
10
11 enum State {
12   MAIN_MENU,
13   PRACTICE_MODE,
14   IoT_MODE,
15   SELECT_DIFFICULTY,
16   EASY_QUESTION,
17   AVERAGE_QUESTION,
18   DIFFICULT_QUESTION,
19   QUESTION_OVER,
20   TRY_AGAIN
21 };
22
23 State currentState = MAIN_MENU;
24 int menuSelection = 0;
25 int difficultySelection = 0;
26 unsigned long currentMillis = 0;
27 unsigned long previousMillis = 0;
28 const long interval = 1000;
29 int questionTimer = 0;
30 int maxTime = 15;
31
32
33 int EasyTimer = 60;
34 int AverageTimer = 180;
35 int DifficultTimer = 300;
36
37 String questionsEasy[] = {"(A AND (A + D) = X, If A=0;D=1", "AB" + BC" = X, If A=0;C=1", "(A'D AND (AB)' = X, If A=0;B=0", "AB'D + B'D' = X, If A=1;B=0=0", "(C' + D) AND (B'C') = X, If B=0;C=0=1"};
38 String questionsAverage[] = {"((AB) + (BC)) AND (C + E)' = X, If B=0;C=0=1", "A'DE + (A' AND (AB))' + DE = X, If A=0;B=0=1", "(A'W'D) + ((D + E)' AND (AB)) + X, If A=0;B=0=1", "[(AB + C) AND (B'D + C'E)] + (AB + C)' = X, If A=0;B=0=1", "A'DE + (A' AND (AB))' + DE = X, If A=0;B=0=1"};

```

```

esp32_practice_mode | Arduino IDE 2.2.0
File Edit Sketch Tools Help
ESP32-WROOM-0A Mod

esp32_practice_mode.ino
140 default:
141   break;
142 }
143
144 void displayFlowingMessage(const String &message) {
145   lcd.clear();
146   for (int i = 0; i < message.length() + 20; i++) {
147     lcd.clear();
148     lcd.setCursor(10 - i, 0);
149     lcd.print(message);
150     delay(200);
151   }
152   lcd.clear();
153 }
154
155 void typeWriteEffect(String text, int row, int delayTime) {
156   lcd.setCursor(0, row);
157   for (int i = 0; i < text.length(); i++) {
158     lcd.print(text[i]);
159     delay(delayTime);
160   }
161 }
162
163 void displayMainMenu() {
164   lcd.clear();
165   lcd.setCursor(0, 0);
166   lcd.print("Choose Mode");
167   lcd.setCursor(0, 1);
168   lcd.print((menuSelection == 0) ? "> Training Mode" : " Training Mode");
169   lcd.setCursor(0, 2);
170   lcd.print((menuSelection == 1) ? "> IoT Mode" : " IoT Mode");
171 }
172
173 void handleSelectButton() {
174   switch (currentState) {

```

```

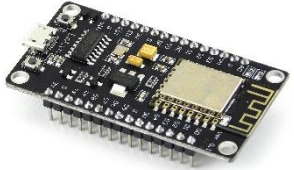
esp3206 | Arduino IDE 2.2.0
File Edit Sketch Tools Help
Generic ESP8266 Module

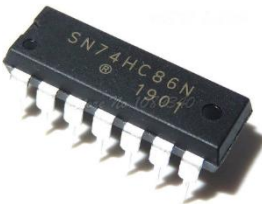
esp3206.ino
1 #include <ESP8266WiFi.h>
2 #include <ESP8266WebServer.h>
3 #include <Wire.h>
4 #include <LiquidCrystal_I2C.h>
5
6 const char* ssid = "Smart Logic Board";
7 const char* password = "ECEL000KIT";
8
9 const int buzzerPin = 32;
10 ESP8266WebServer server(80);
11 LiquidCrystal_I2C lcd(0x27, 20, 4);
12
13 int numQuestions = 0;
14 String problems[10];
15 int timerDuration[10];
16 String answers[10];
17
18 int currentQuestion = 0;
19 unsigned long startTime = 0;
20 unsigned long questionStartTime = 0;
21 bool displayProblem = false;
22 bool displayReCountdown = false;
23 bool displayCountdown = false;
24 bool displayTimeUp = false;
25 bool displayAnswer = false;
26
27 void handleRoot() {
28   String html = "<html><head><title>Smart IoT0 Training Kit</title><style>"
29     "body { background-color: cyan; display: flex; flex-direction: column; justify-content: center; align-items: center; height: 100vh; margin: 0; text-align: center; font-family: Arial, sans-serif; }"
30     "h1 { margin-top: 20px; margin-bottom: 10px; color: darkblue; font-weight: bold; font-size: 48px; }"
31     "h2 { margin-bottom: 20px; color: darkblue; font-size: 24px; }"
32     "form { display: flex; flex-direction: column; align-items: center; }"
33     "label { font-size: 20px; margin: 10px 0; color: darkblue; }"
34     "input[type='number'], input[type='text'] { padding: 15px; width: 300px; font-size: 20px; margin: 10px 0; border-radius: 5px; border: 1px solid #ccc; }"
35     "input[type='submit'] { padding: 15px 20px; font-size: 20px; cursor: pointer; margin-top: 10px; border: none; border-radius: 5px; background-color: #00AEEF; color: white; }"
36     "</style></head><body>";

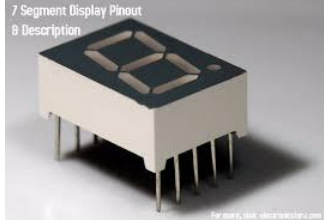
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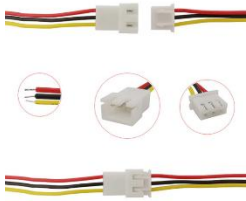
Appendix H

Pictorial Presentation of Different Hardware Parts and their Functions of the Smart Lobo Training Kit

COMPONENT NAME	PICTURE OF COMPONENT	DESCRIPTION FUNCTION
ESP8266	 https://circuit.rocks/cdn/shop/products/nodemcu-v3-esp8266-12e-development-board-ch340-1442.jpg?v=1689674249	Enables microcontrollers to connect to 2.4 GHz Wi-Fi networks using the full TCP/IP stack, making it versatile for IoT applications with data handling and GPIO control capabilities.
ESP32	 https://www.az-delivery.de/cdn/shop/products/esp32-nodemcu-module-wlan-wifi-development-	A powerful chip for providing Wi-Fi and optional Bluetooth connectivity, used extensively in embedded and IoT devices, offering high performance and versatility in data transmission.
SN74HC08	 https://encrypted-tbn0.gstatic.com/images?q=tbn:	A high-speed quad 2-input AND gate IC, ideal for performing logical AND operations in digital circuits, data manipulation, and arithmetic functions.
SN74HC00	 https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTpGr5SvKXXB-pMltvFaZ8PVvuVxdeah1H-	A quad 2-input NAND gate IC that efficiently handles NAND operations, widely used in digital circuits for essential logic functions and signal processing.

SN74HC04	 https://encrypted-tbn0.gstatic.com/images?q=tbn:AND9GcTH1NskXRMKrkaHrSrSL--daDWLG41wnEzow&s	hex inverter IC that inverts input signals, featuring six independent NOT gates, perfect for signal inversion tasks and logic operations in digital circuits.
SN74HC02	 https://encrypted-tbn0.gstatic.com/images?q=tbn:AND9GcR6xoQpCRBrJ6BQPHfA3-4OgtKR-Jib51wKxA&s	A quad 2-input NOR gate IC that handles high-speed NOR operations, making it useful for data processing and various digital logic applications.
SN74HC32	 https://encrypted-tbn0.gstatic.com/images?q=tbn:AND9GcQwVEfRPdgdfS5cDuHv3N9PACzfNTuQ	A quad 2-input OR gate IC designed for reliable OR operations in digital logic circuits, frequently used for data processing and circuit design.
SN74HC86	 https://m.media-amazon.com/images/I/51e7gCE3iuL._AC_UF1000,1000_QL80_.jpg	A quad 2-input XOR gate IC for exclusive OR operations, commonly applied in error d

CD4077	 https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQn-xPxA4EzgmKJ-	A dual 2-input XNOR gate IC that performs exclusive NOR operations, ideal for applications requiring data comparison and arithmetic operations with low power consumption.
LCD I2C 20X4	 https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQ_gugk9aRjYpxWz39vTaxsz20mfPgFo	A 20-character by 4-line liquid crystal display using I2C communication, simplifying wiring for embedded systems and DIY electronics, perfect for compact user interfaces.
1 INCH 1 DIGIT 7 SEGMENT DISPLAY COMMON CATHODEE	 https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTgEb8J64IQZUASiyNdzSTCRhddO7hhxt	Displays decimal digits (0-9) with seven segments, easily controlled by microcontrollers for applications like digital meters and clocks.

3 PIN ROCKER SWITCH	 https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRh0hyqq9296yYdDC2JPDeFflaXt7eBhS	A reliable electrical switch that toggles between on/off positions, commonly used in appliances and devices to manage current flow in circuits.
BUZZER	 https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRUfiRKNpR4ZrrU_f4y48hnoWdqZ7D9v	An electronic sound emitter used for alarms or notifications, available in passive (external signal required) and active (self-sounding) types, easy to integrate into circuits.
CD4511	 https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTrHoLVQmJuVrcO_1aLrCtVcrs52uOPR6	A BCD to 7-segment decoder/driver IC that converts binary input to decimal output for driving 7-segment displays, ideal for calculators and digital readouts.
JST CONNECTOR 3 PINS	 https://www.railwayscenics.com/images/electrical/3way-miniature-connectors.jpg	A small, secure connector used for reliable electrical connections in electronics, ideal for battery packs and lightweight projects.

PUSH BUTTON	 https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcR2jwnbYt1xONqGyIKRAyWLboexAPb1S	A simple mechanical switch that completes or interrupts a circuit when pressed, commonly used to trigger functions like turning devices on/off or resetting.
POWER SUPPLY CENTER NEGATIVE	 https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcS3E66r7MkLRSQ7aEotFOcBySsCvL8Ti3	A power adapter configured with negative voltage at the center pin, ensuring proper polarity for devices that require this specific setup.

Appendix I

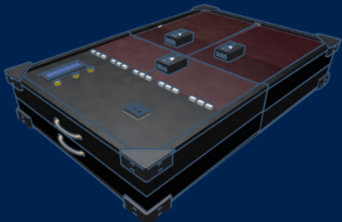
Bill of Materials

QUANTITY	UNIT	DESCRIPTION	PRICE(PHP)	TOTAL PRICE(PHP)
10	piece	A3 Sintra Board	35.8	358
1	piece	ESP32	204	204
1	piece	ESP8266	113	113
5	piece	SN74HC08	5	25
5	piece	SN74HC04	5	25
5	piece	SN74HC02	5	25
5	piece	SN74HC32	5	25
5	piece	SN74HC00	5	25
5	piece	SN74HC86	5	25
2	piece	CD4077	25	50
1	piece	CD4511	25	25
1	piece	LCD Display with I2C	250	250
1	piece	7 Segment Display Common Cathode	75	75
7	piece	3 Pin Rocker Switch	16	112
1	piece	Buzzer	29	29
42	piece	JST Connector 3 Pins	6	252
3	piece	Push Button	18	54
1	piece	9V Power Supply Center Negative	50	50
10	piece	5V LM7805 Regulator	9	90
Total Cost				1,812

Appendix J

User Manual

SMART LoBo TRAINING KIT: USER MANUAL





Getting Started:

- 1. Open Up the Logic Board**
Carefully expand the board to access all parts.
- 2. Power Connection**
Use the kit's dedicated power supply, ensuring the power jack is correctly connected according to the polarity indicator.

- 3. Turn on the device**
Flip the switch on the side of the device to power it on.
- 4. Selecting a Mode**
Choose between **Training Mode** or **IoT Mode** by flipping the Mode Selector Switch, located on the side of the board.

- 5. Creating the Logic Circuit**
Place the logic blocks and connect inputs/outputs based on the logic expression. Use the switch to toggle inputs between "0" and "1."
- 6. Output Identifier**
The output of the circuit is displayed on the seven-segment display, showing whether the circuit is 1 or 0.

Precautionary Measures

- When connecting the logic blocks, ensure that **all inputs** are properly connected and none are left unconnected. This will prevent interference or floating inputs, which can cause unpredictable changes in the output.
- Do not** connect multiple outputs at a time. Only one output must be connected to avoid damaging the components or getting incorrect outcomes.

INTERFACE OVERVIEW

The Logic Training Kit interface is designed for easy navigation and control, using the following buttons:

- Select:** Choose an option
- Option:** Access additional settings.
- Back:** Return to the previous menu.

When powered on, the LCD displays the available modes. Use the buttons to select between Practice Mode and IoT Mode.

TROUBLESHOOTING

If the LCD display shows no content or not functioning properly, follow these steps:

- Flip the board over and open the compartment containing the electrical connections.
- Press the Reset button on both microcontrollers, labeled **EN** and **RST**.

This should restore the display and resolve any issues with the LCD.



TRAINING MODE

In Training Mode, users can work with predefined logic equations to enhance learning experience.

Features include:

- Timed Practice:** Work on logic problems with difficulty levels:
 - Easy
 - Average
 - Difficult
- The logic equation is displayed for 25 seconds before the timer starts, so it's advised to take note of the equation.
- As the questions are randomized, repeated problems may occur.

Note: Timer duration varies by difficulty, and the buzzer sounds when time expires.

IOT MODE

In IoT Mode, users can customize their own logic exercises through a mobile device or computer.

Features include:

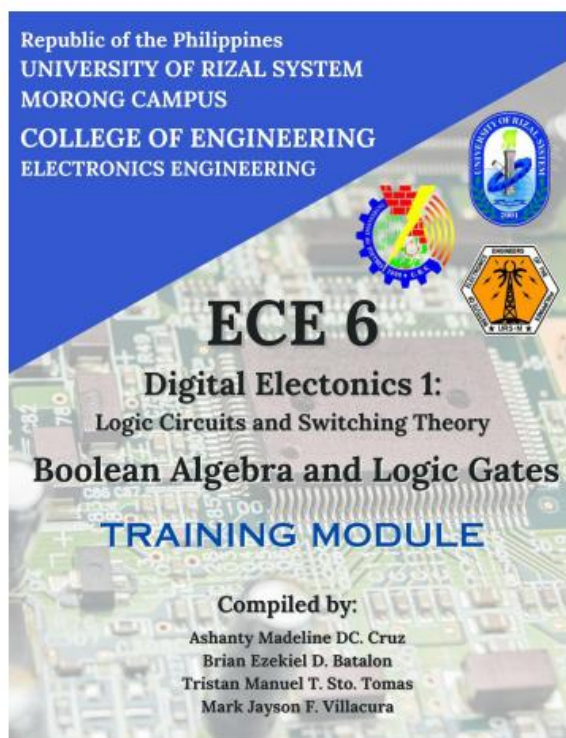
- Custom Logic Expression:** User can input expression to the LCD display.
- Network Host:** The microcontroller will act as a network host, allowing devices to connect to a dedicated Wi-Fi network.
- Web Server Access:** Once connected to the network, devices can access the web server to input custom equations and set time limits.

To do this, connect to the ESP8266 Wi-Fi module named **"Smart Logic Board"** (password: **ECELOBOKIT**) and open a web browser to **192.168.4.1**.

Final Note: Follow safety precautions for proper use and durability. Enjoy building logic circuits!

Appendix K

Photos of Training Module



Welcome to the SMART LoBo Training Kit Module!

Technology has transformed the way we learn, making knowledge more accessible than ever. In today's world, understanding digital logic is not just an advantage—it's essential. This training module, featuring the hands-on SMART LoBo Training Kit, is designed to make your learning journey engaging and effective, providing you with the resources and activities needed to explore digital logic circuits in depth.

The SMART LoBo Training Kit is a powerful digital logic trainer that brings theoretical concepts to life. By simulating computer circuitry with various ICs, this kit allows you to easily implement logic circuits. Like Logisim, a digital logic simulation software, it enables you to set logic level inputs, connect gates, and observe logic level outputs in real time. The trainer includes a power supply, logic blocks of various gates, a seven-segment display for output indication, logic level switches, and 3-pin JST connectors. It is enclosed in a plastic case with a latch and a handle for added convenience and safety.

As the creators of this module, we are committed to ensuring you gain valuable insights and skills from each section. If you have any questions or need further assistance, please don't hesitate to reach out using the contact options below.

Happy learning, and enjoy discovering the world of digital logic!

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LESSON 1: LOGIC GATES FUNDAMENTALS

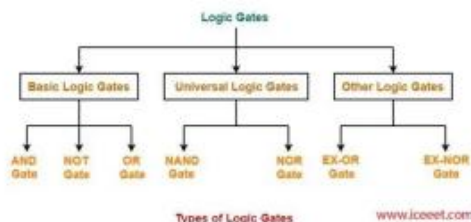
Learning Objectives:

At the end of this lesson, the students are expected to:

1. Know the Logic Gates and its Symbols
2. Know the equivalent truth table of logic gates
3. Understand Logic Gates and its Operation
4. Familiarize with Logic Gates Integrated Circuits (ICs)

INTRODUCTION:

Logic gates perform basic logical functions and are the fundamental building blocks of digital integrated circuits. It is an electronic circuit having one or more than one input and only one output. Most logic gates take an input of two binary values, and output a single value of a 1 or 0. Some circuits may have only a few logic gates, while others, such as microprocessors, may have millions of them.



Representation of Logic gates

There are three different, but equally powerful, notational methods for describing the behavior of gates and circuits.

1. **Logic Diagram:** A graphical representation of a circuit. Each type of gate is represented by a specific graphical symbol.

1.1 OR GATE

I. Objectives

1. To examine the characteristics and operation of a TTL OR Gate.
2. To construct a logic circuit of an OR Gate with an expanded input using 2-input TTL OR Gates.

II. Equipment and Materials

- Digital Logic Trainer Board
- 4 pcs OR Gate logic blocks (7432 IC, a quad 2-input OR Gate)
- 3-pin connector wires (JST connectors)

III. Introductory Information

The logic operation performed by the OR gate is logic addition, also known as the OR operation or function. The OR gate has two or more inputs and a single output. Table 1 shows the ANSI & IEC logic symbols for an OR gate with two input variables A and B. The output of this gate is expressed in Boolean algebra notation as $A+B$. This notation is read as "A or B", not as "A plus B".

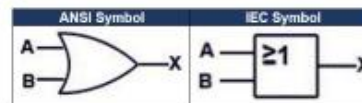


Table 1. Logic symbols for a two-input OR gate

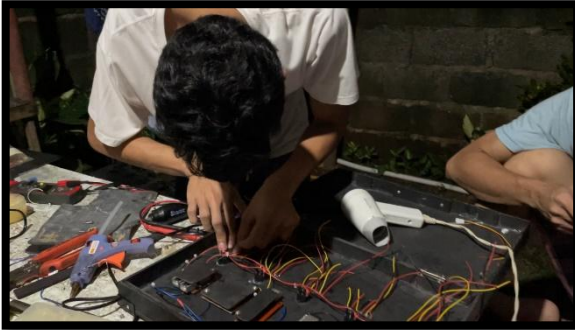
For a 2-input OR gate:

- Output (X) is HIGH (1) when either A or B is HIGH, or when both A and B are HIGH.
- Output (X) is LOW (0) only when both A and B are LOW.

The OR gate determines when certain conditions are met simultaneously. The OR gate is an any-or-all gate because a 1 at any input gives rise to a 1 at the output. The HIGH (1) level is the prime or active output level for the OR Gate.

Appendix L

Documentation



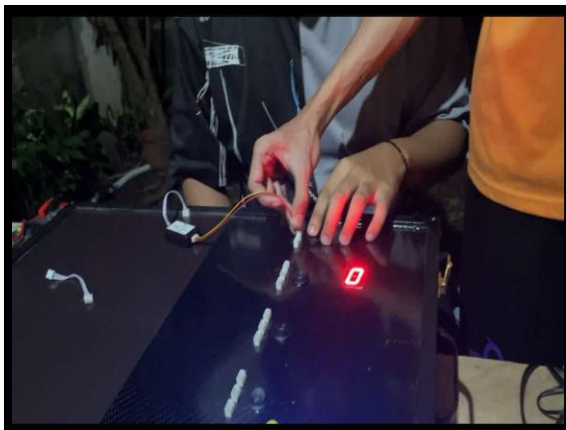
Wiring of the device



Breadboarding



Chassis Construction



Testing of the device



Conduct of Survey



Final Defense

CURRICULUM VITAE

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Teresa National High School, Science Technology Engineering and Mathematics (STEM)

Junior High School (2015-2019)

Teresa National High School

Elementary (2009 - 2015)

St. Rose of Lima Montessori School

WORKSHOP/SEMINARS/TRAININGS

IECEP-URSMSC Technical Seminar (2024)

E-Connect: Establishing Transmission of Intelligence from Professionals to Future Innovators

Asia Open RAN Academy, DICT, USAID Webinar (2023)

Cloud or Edge: Understanding the Network Environment.

IECEP-RSC Webinar (2023)

Exploring Ideas and Innovating Techniques in Forming Technologically Inclined Future

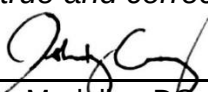
IECEP-URSMSC Technical Seminar (2022)

Electronics Manufacturing: Conceptualizing ECE fields through professional context.

IECEP-URSMSC Technical Seminar (2022)

Radio Broadcasting: Conceptualizing ECE fields through professional context.

I hereby certify that the above information given are true and correct as to the best of my knowledge.


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Arduino Workshop 2022, Level Up: ECE for Excellence

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